

Design of A Multiple Channel Auto-Tuner

Lewis Thompson, W5IFQ, P.E.

6 OCT 2026

Purpose

- To outline the design criteria that I and my customer used to improve the performance of his remote, HF MARS (public service) radio station.
- As you will see, we eventually arrived at a solution that required the design and construction of multi-channel wideband auto tuner system.

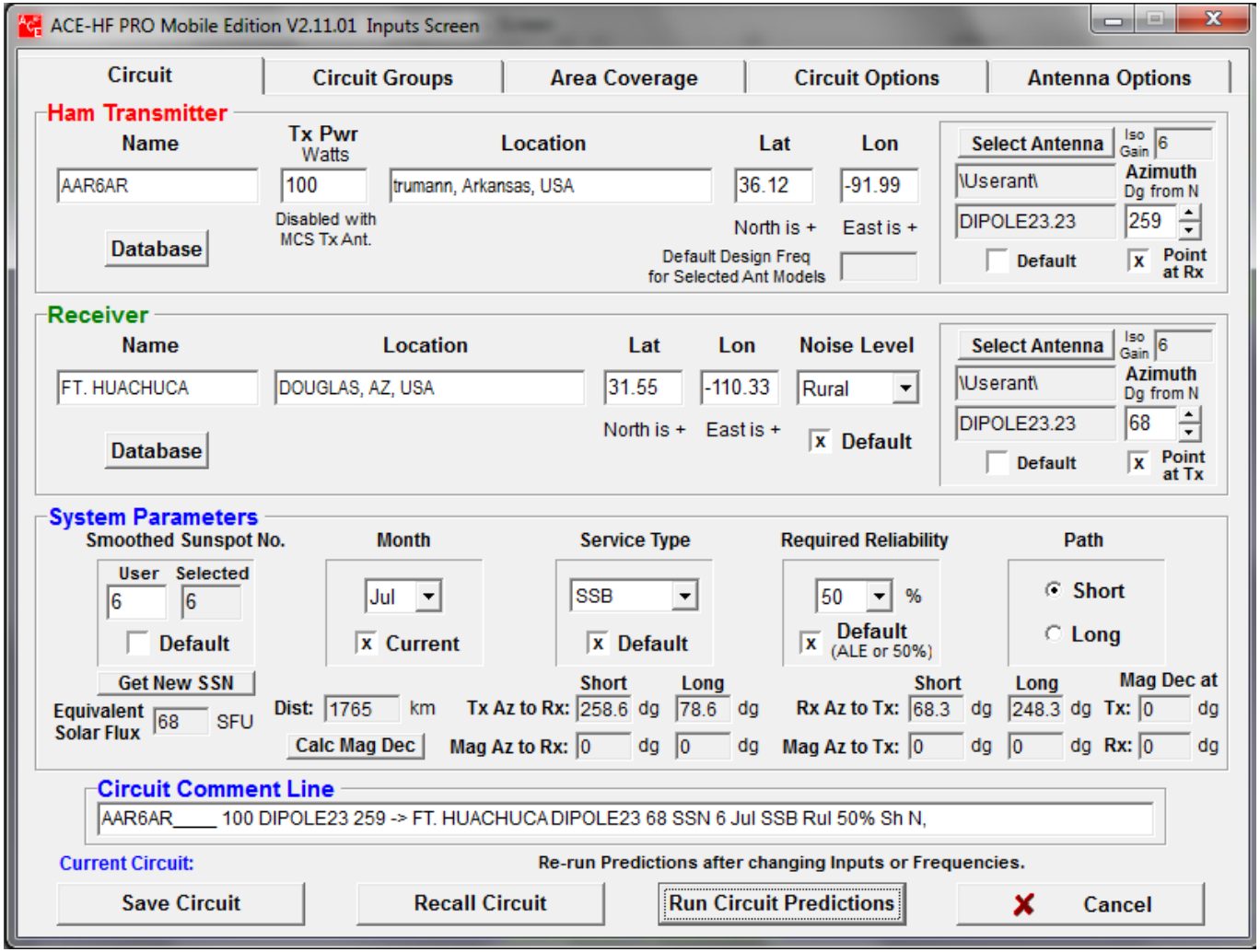
Introduction

- This Multi-channel Auto Tuner allows remote selection and auto-tuning of up to 5 separate, non-resonant HF antennas.
- This can include both balanced dipoles and single-ended verticals and long-wires.
- This tuner found low SWR solutions on all Army MARS frequencies for the Cobra Ultralight Junior and Senior and a Netcom 66/44 antennas at 50 ft. heights.

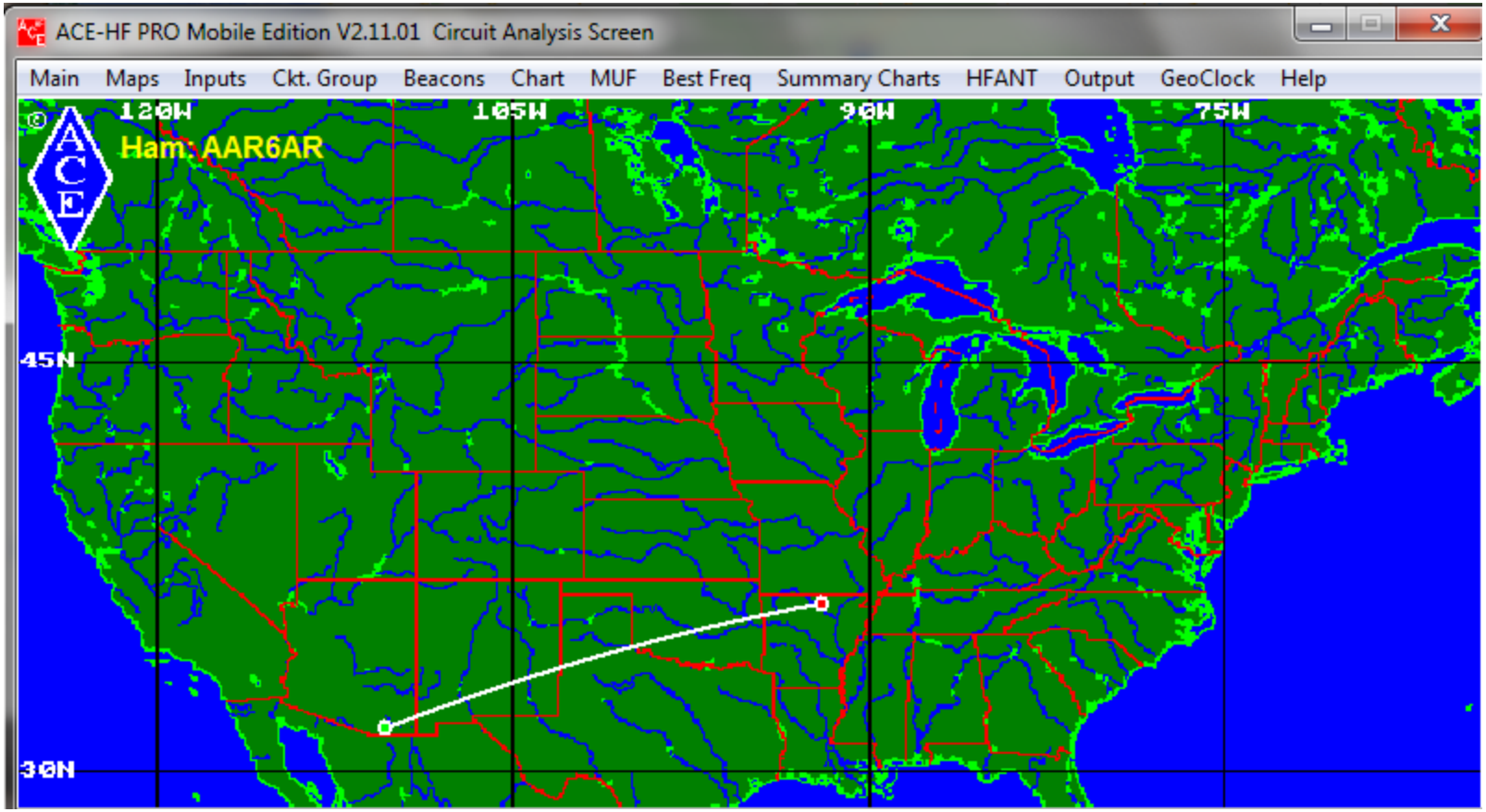
Antenna Performance Requirements

- My Army MARS Customer, Tim, lives in NE Arkansas and wanted good HF communications to the following locations:
  - NVIS Arkansas area (250 mile radius, TOA - 50˚)
  - Austin, TX (523 mi heading - 221˚ T, TOA - 30˚)
  - Louisiana (500 + mi, heading - 180˚T, TOF - 30˚)
  - Fort Huachuca, AZ (Army Netcom) (1096 mi heading - 259˚T, TOA - 20˚)
- To achieve this required variety of ranges and headings, a number of dipoles at different heights and heading must be deployed.
- Further, Army MARS uses a wide range of Military HF frequencies from 3 MHz to 18 MHz in approximately 1 MHz steps.
- A VOACAP based propagation program, ACE-HF, was used to determine Range, Headings and TOA's for the required circuits listed above.
- An example for Ft. Huachuca is shown on pages 4 – 7.

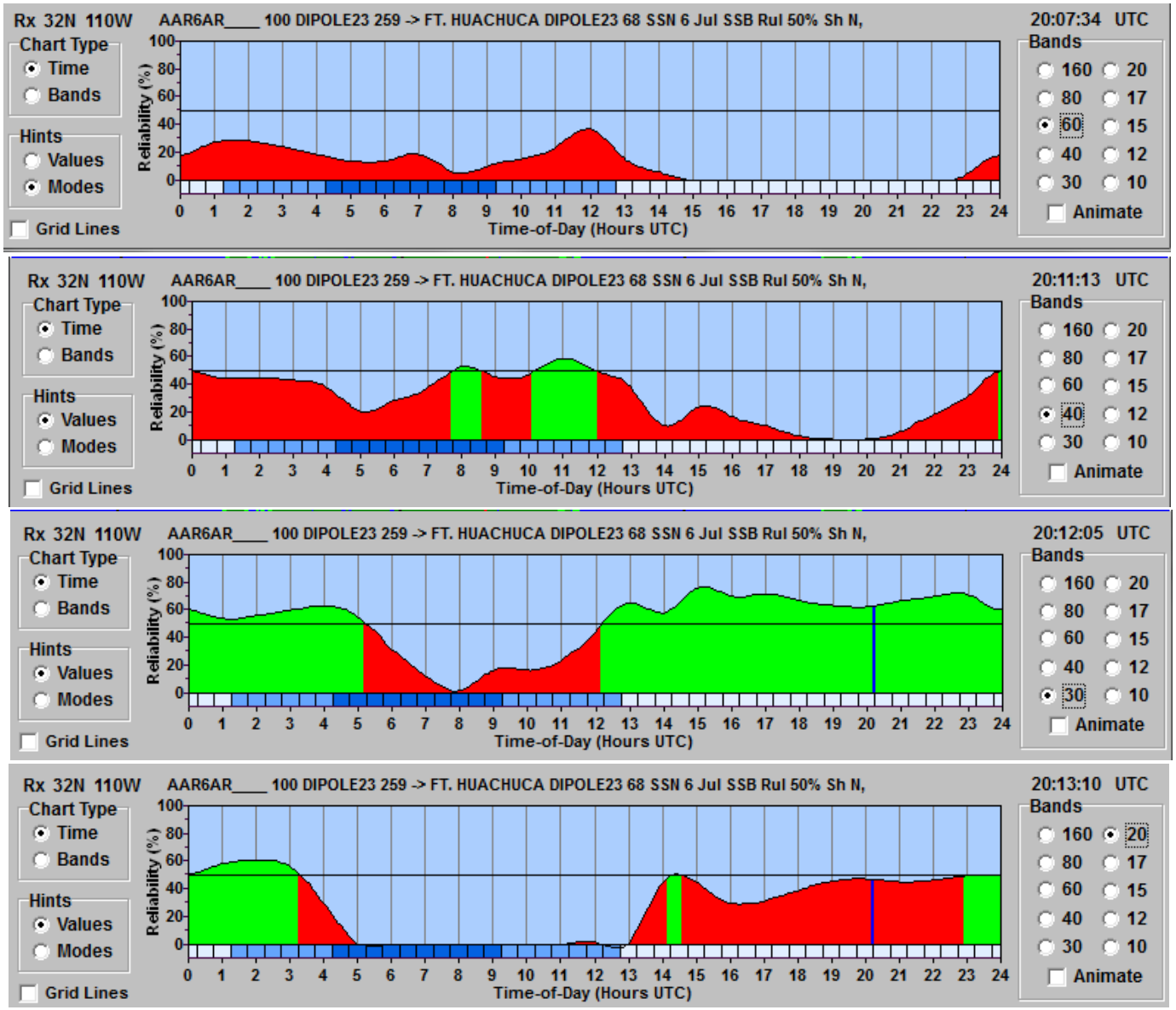
Ft. Huachuca ACE-HF Setup



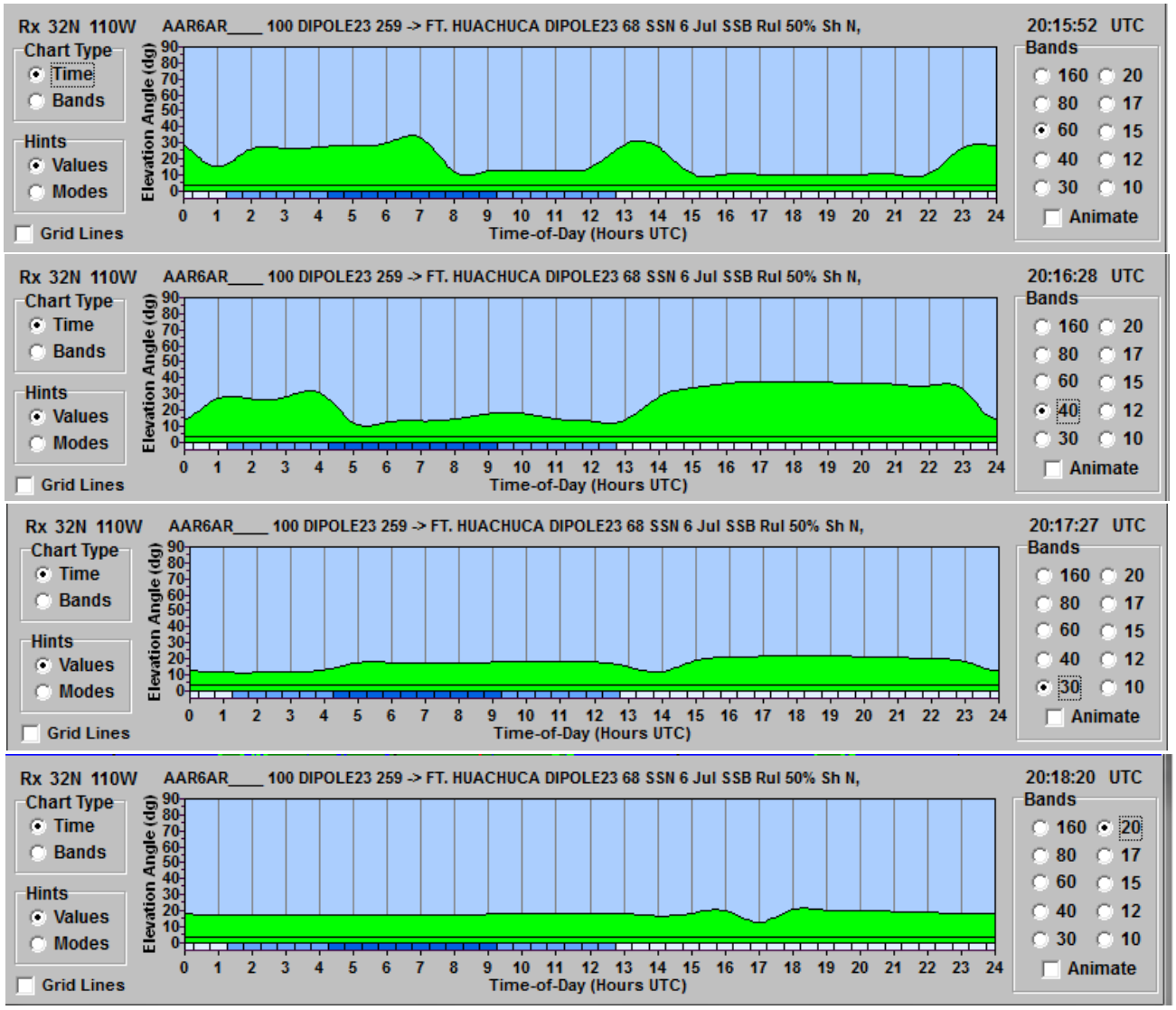
MAP – Ft. Huachuca



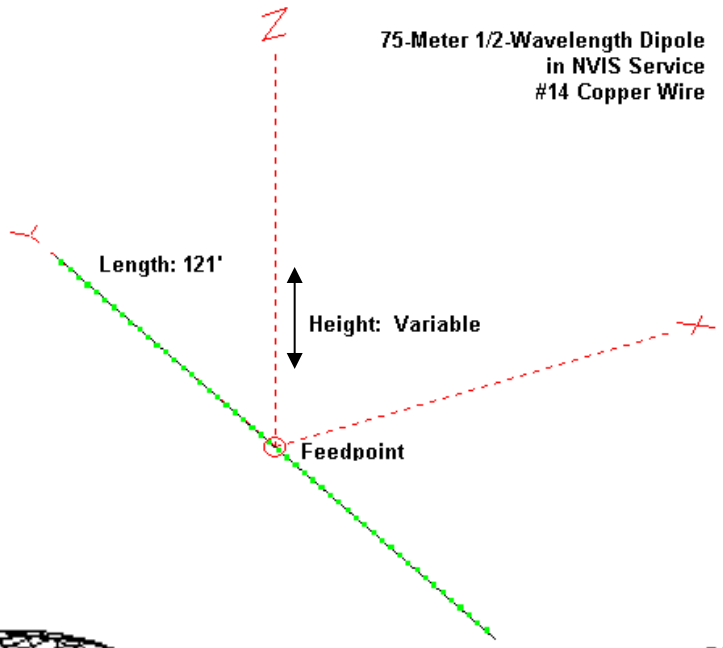
Reliability – Ft. Huachuca Circuit



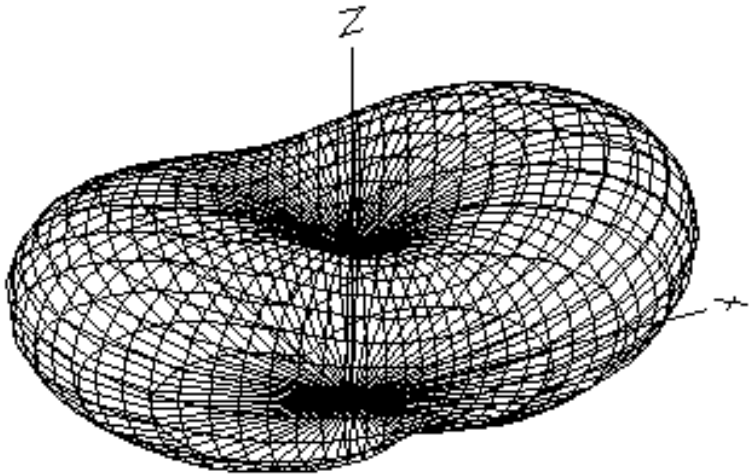
Take-Off Angles – Ft. Huachuca



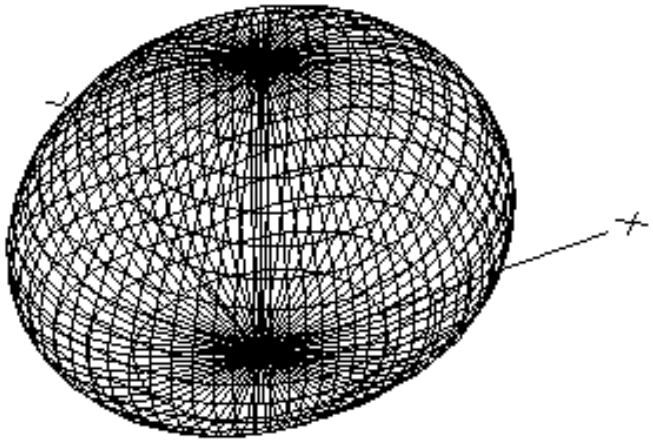
Dipole Height Versus Directivity



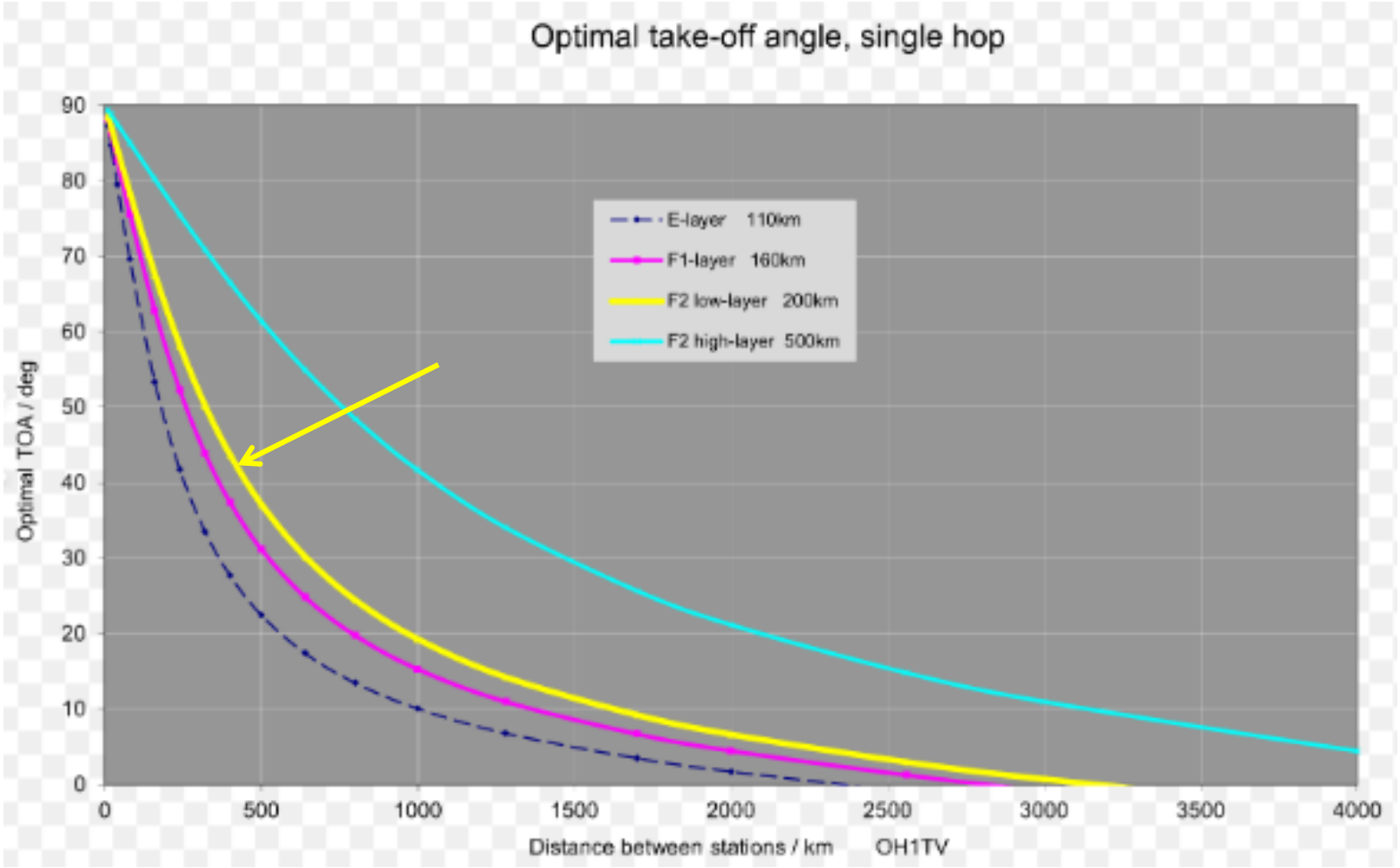
Height = ½ λ



Height = 0.12 λ



TOA Versus Range



Dipole Height versus Range

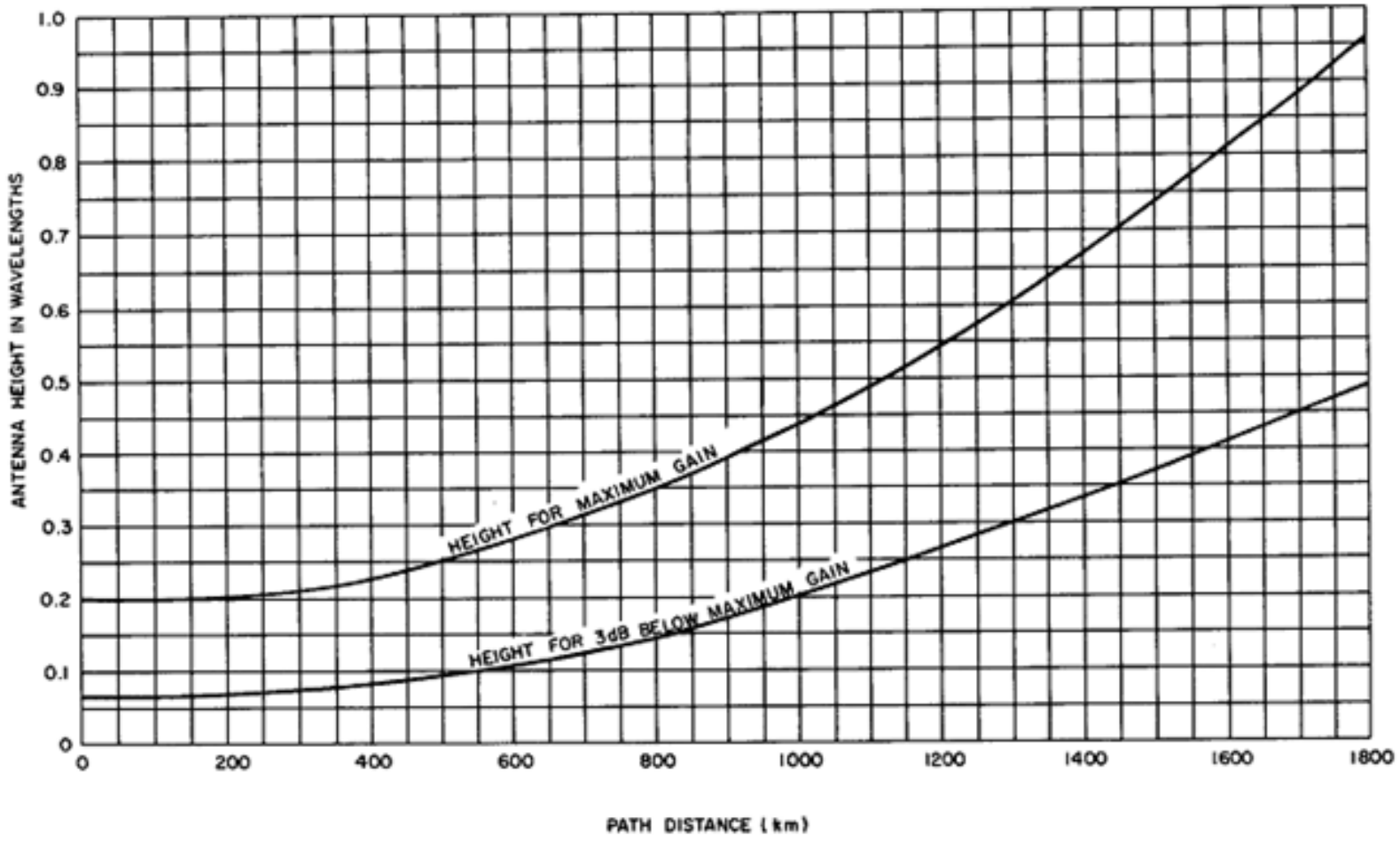


Figure 5–3. Approximate height of half-wave dipole for best F2 layer propagation.

Frequency versus Wavelength

- When designing antennas, the length and height in wavelengths determines the antenna characteristics. It is easier to work in wavelengths and then convert to actual dimensions at the design frequency(s):

  λ (wavelength, ft.) = 983.6/f (frequency, MHz)

- A convenient lookup table follows.

Frequency to Wavelength Chart

Frequency (kHz)		Wavelength (Meters)		Wavelength (Feet)		1/2 Wavelength (Feet)		1/4 Wavelength (Feet)		1/3 Wavelength (Feet)		1/10 Wavelength (Feet)	
2,194.00	2,495.00	136.7	120.2	448.6	394.5	224.3	197.2	112.2	98.6	149.5	131.5	44.9	39.4
2,505.00	2,850.00	119.8	105.3	392.9	345.3	196.5	172.7	98.2	86.3	131.0	115.1	39.3	34.5
3,155.00	3,400.00	95.1	88.2	312.0	289.5	156.0	144.7	78.0	72.4	104.0	96.5	31.2	28.9
4,000.00	4,063.00	75.0	73.8	246.1	242.2	123.0	121.1	61.5	60.6	82.0	80.7	24.6	24.2
4,438.00	4,650.00	67.6	64.5	221.8	211.7	110.9	105.8	55.4	52.9	73.9	70.6	22.2	21.2
4,750.00	4,995.00	63.2	60.1	207.2	197.0	103.6	98.5	51.8	49.3	69.1	65.7	20.7	19.7
5,005.00	5,450.00	59.9	55.0	196.7	180.6	98.3	90.3	49.2	45.1	65.6	60.2	19.7	18.1
5,730.00	5,950.00	52.4	50.4	171.8	165.4	85.9	82.7	42.9	41.4	57.3	55.1	17.2	16.5
6,765.00	7,000.00	44.3	42.9	145.5	140.6	72.7	70.3	36.4	35.2	48.5	46.9	14.5	14.1
7,300.00	8,195.00	41.1	36.6	134.8	120.1	67.4	60.1	33.7	30.0	44.9	40.0	13.5	12.0
9,040.00	9,500.00	33.2	31.6	108.9	103.6	54.4	51.8	27.2	25.9				
9,900.00	9,995.00	30.3	30.0	99.4	98.5	49.7	49.2	24.9	24.6				
10,150.00	11,175.00	29.6	26.8	97.0	88.1	48.5	44.0	24.2	22.0				
11,400.00	11,650.00	26.3	25.8	86.3	84.5	43.2	42.2	21.6	21.1				
12,050.00	12,230.00	24.9	24.5	81.7	80.5	40.8	40.2	20.4	20.1				
13,410.00	13,600.00	22.4	22.1	73.4	72.4	36.7	36.2	18.3	18.1				
13,800.00	14,000.00	21.7	21.4	71.3	70.3	35.7	35.2	17.8	17.6				
14,350.00	14,990.00	20.9	20.0	68.6	65.7	34.3	32.8	17.1	16.4				
15,600.00	16,360.00	19.2	18.3	63.1	60.2	31.5	30.1	15.8	15.0				
17,410.00	17,550.00	17.2	17.1	56.5	56.1	28.3	28.0	14.1	14.0				
18,030.00	18,068.00	16.6	16.6	54.6	54.5	27.3	27.2	13.6	13.6				
18,168.00	18,780.00	16.5	16.0	54.2	52.4	27.1	26.2	13.5	13.1				
18,900.00	19,660.00	15.9	15.3	52.1	50.1	26.0	25.0	13.0	12.5				
19,800.00	19,990.00	15.2	15.0	49.7	49.2	24.9	24.6	12.4	12.3				
20,010.00	21,000.00	15.0	14.3	49.2	46.9	24.6	23.4	12.3	11.7				
21,855.00	23,200.00	13.7	12.9	45.0	42.4	22.5	21.2	11.3	10.6				
23,350.00	24,890.00	12.8	12.1	42.2	39.5	21.1	19.8	10.5	9.9				
25,330.00	25,550.00	11.8	11.7	38.9	38.5	19.4	19.3	9.7	9.6				
26,480.00	28,000.00	11.3	10.7	37.2	35.2	18.6	17.6	9.3	8.8				
29,800.00	30,000.00	10.1	10.0	33.0	32.8	16.5	16.4	8.3	8.2				

Minimum NVIS Height

Maximum NVIS Height

Long-Range Height

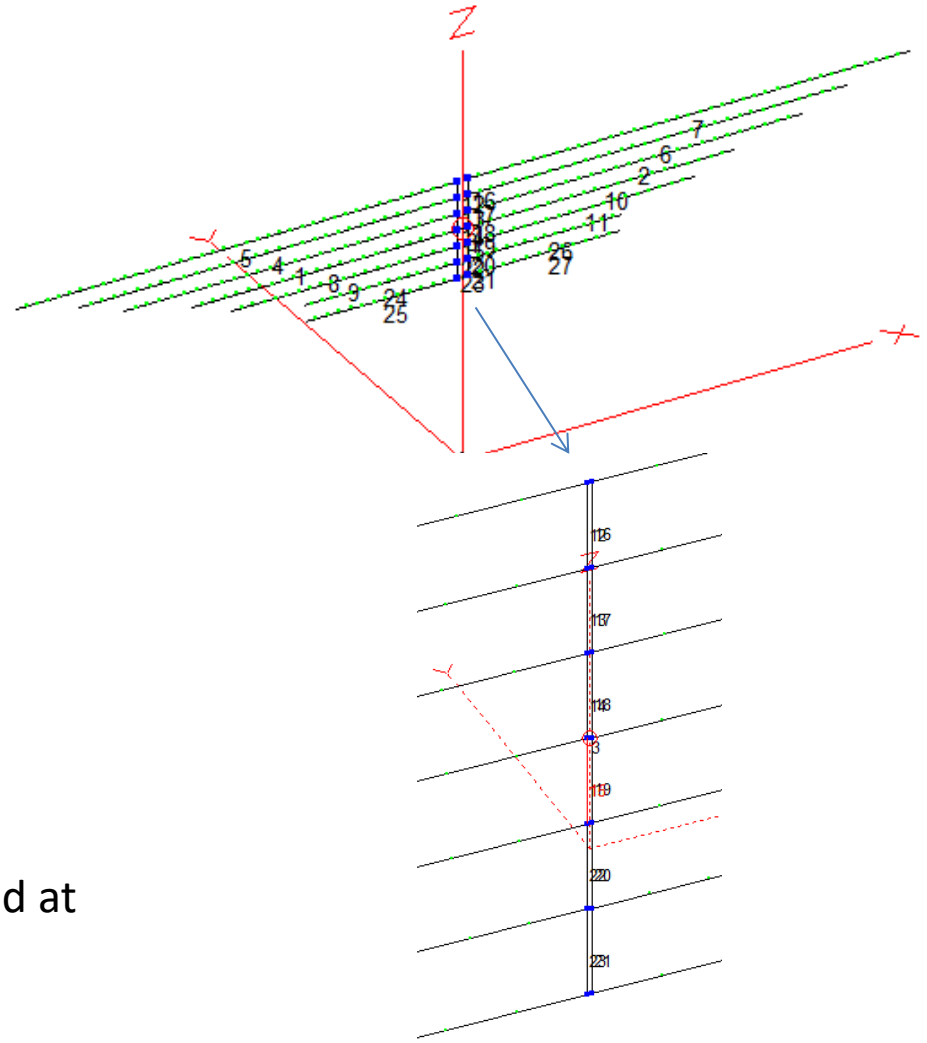
First Attempts At Antenna Requirements

- Vertical Antennas were found to have insufficient gain at 100 watts transmit levels.
- Nested, resonant dipoles at different heights would be very difficult to construct and tune.
- Nested, resonant dipoles at different headings would also be very difficult to construct and tune.

Vertically Spaced, Resonant Dipoles (ALE Station)

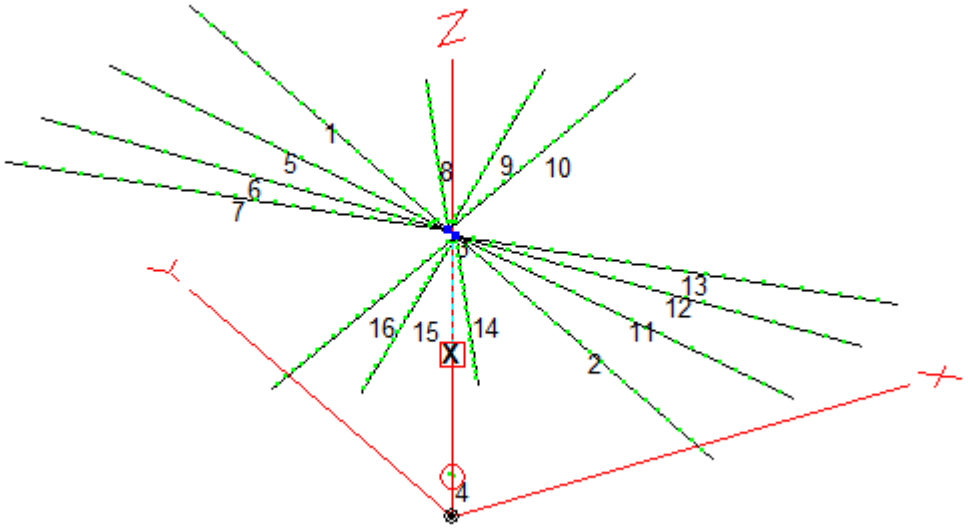
Wires 5, 7 (4.007 MHz) : 56 ft. each
Wires 4, 6: (4.9185 MHz): 48.5 ft. each
Wires 1, 2: (5.763 MHz): 41.5 ft. each
Wires 8, 10 (6.907 MHz): 34.5 ft. each
Wires 9, 11 (7.953 MHz): 28.5 ft. each
Wires 24, 26 (9.9405 MHz): 20 ft. each
Wires 25, 27 (12.147 MHz): 19.8 ft. each
All other wires are for interconnection and source (wire 3).
All wires #12 AWG
Top height – 35 ft.
Horizontal Spacing – 1 inch
Vertical Spacing – 2 ft.

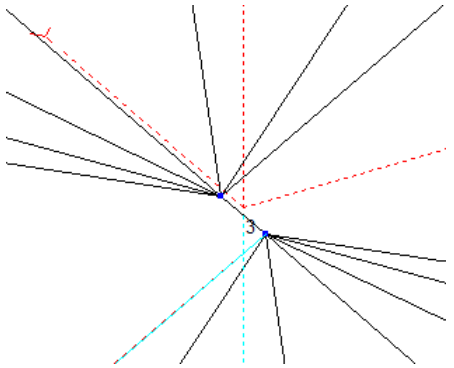
Note - 1:1 current balun should be located at source location.



Horizontally Spaced, Resonant Dipoles

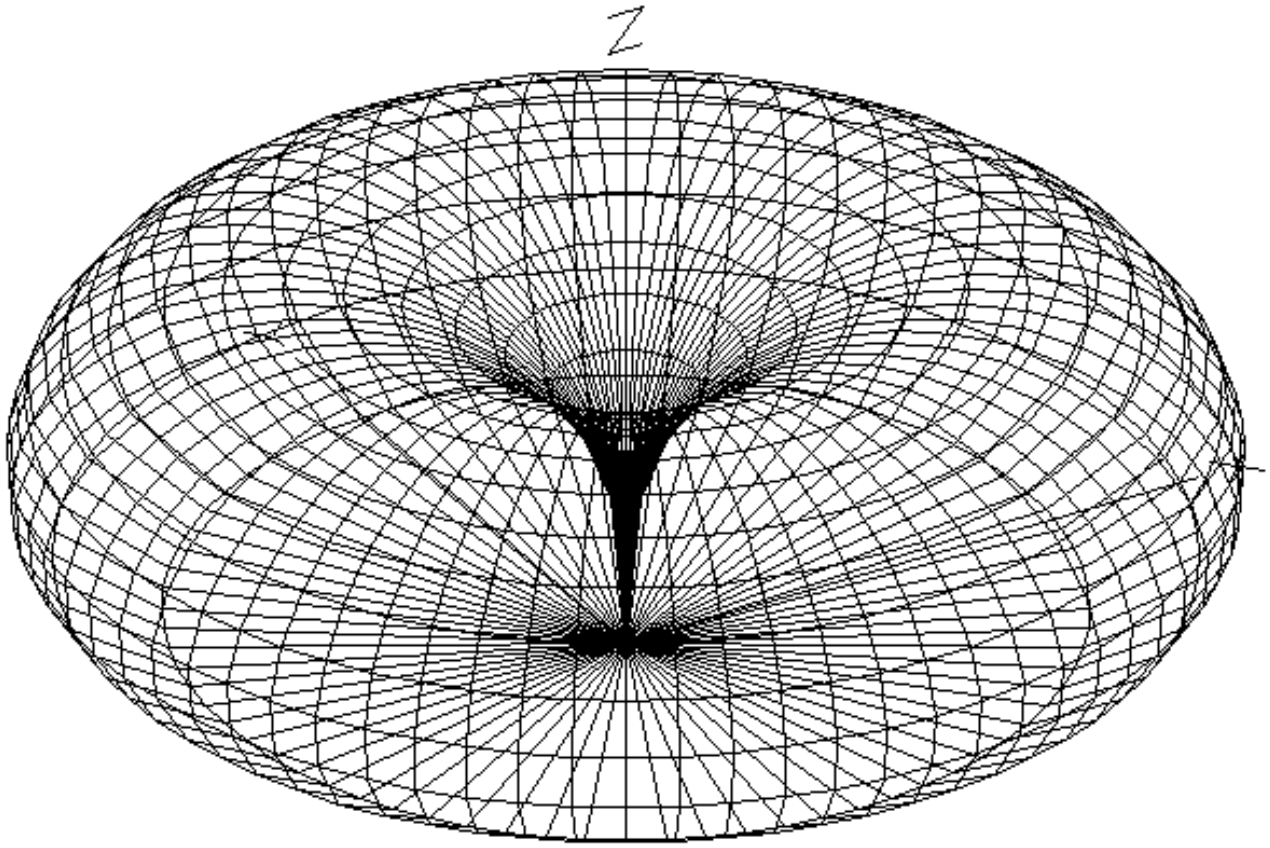
Wires 1,2 (4.007MHz) : 56 ft. each
Wire 3: 1.5 ft. EZNEC driving point
Wires 5, 11: (4.9185 MHz): 49 ft.
Wires 6, 12: (5.763 MHz): 43 ft.
Wires 7, 13 (6.907 MHz): 37 ft.
Wires 8, 14 (7.953 MHz): 29 ft.
Wires 9, 15 (9.9405 MHz): 23.8 ft.
Wires 10, 16 (12.147 MHz): 20 ft.
All wires #14 AWG
All heights are horizontal at 35 ft.





X – 1:1 current balun located at antenna junction
O 4 - This is location for attachment of coaxial cable

Vertical Antenna Pattern (Not suitable for NVIS)



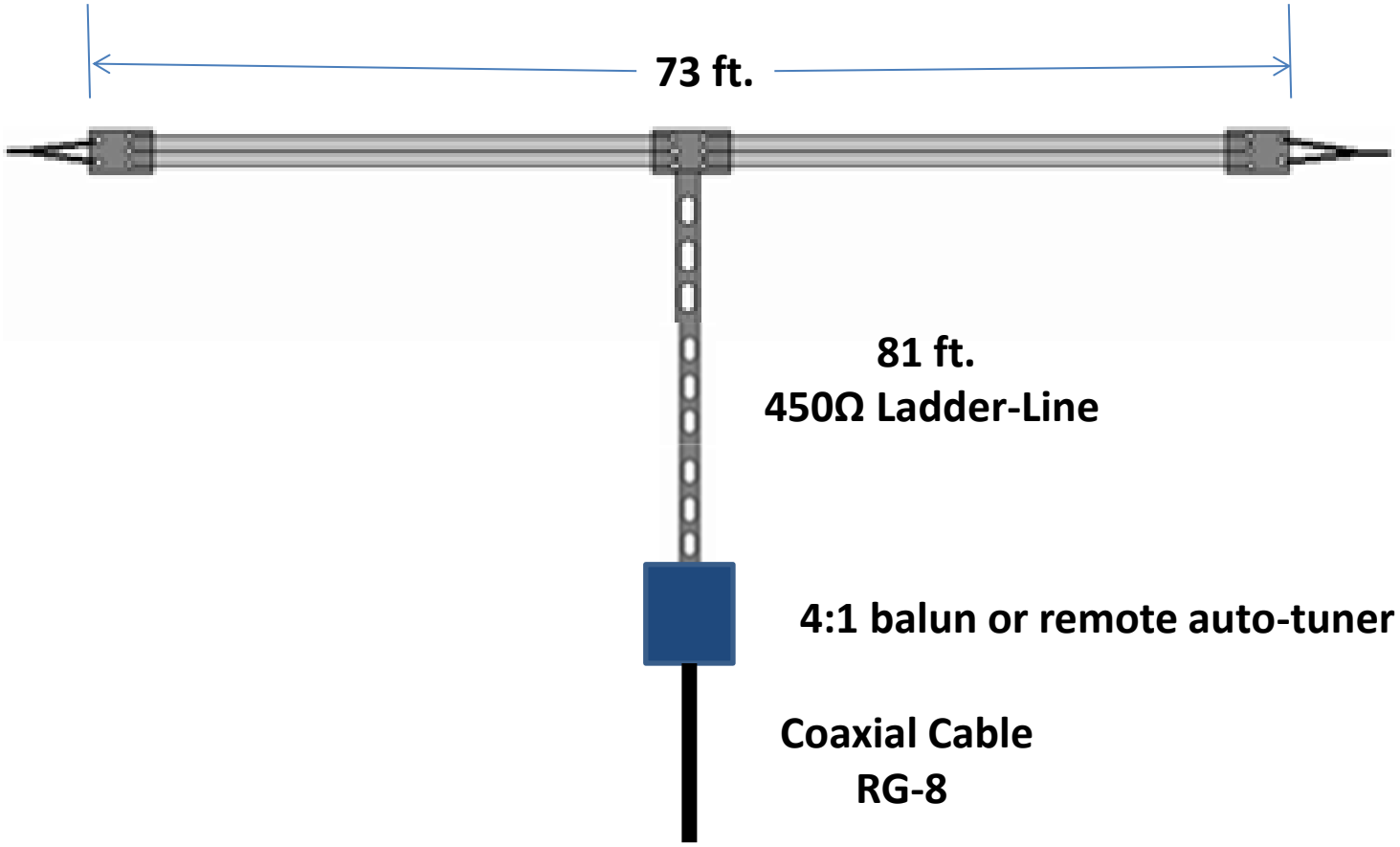
Antenna Decisions

- It quickly became obvious that stacks of resonant dipoles was not the answer.
  - Too many required operating frequencies,
  - A variety of headings,
  - A variety of required heights to achieve TOA's
  - Very difficult to adjust to resonance.
- The many required operating frequencies drove the design to use a few wide-bandwidth dipoles at variety of headings and heights.

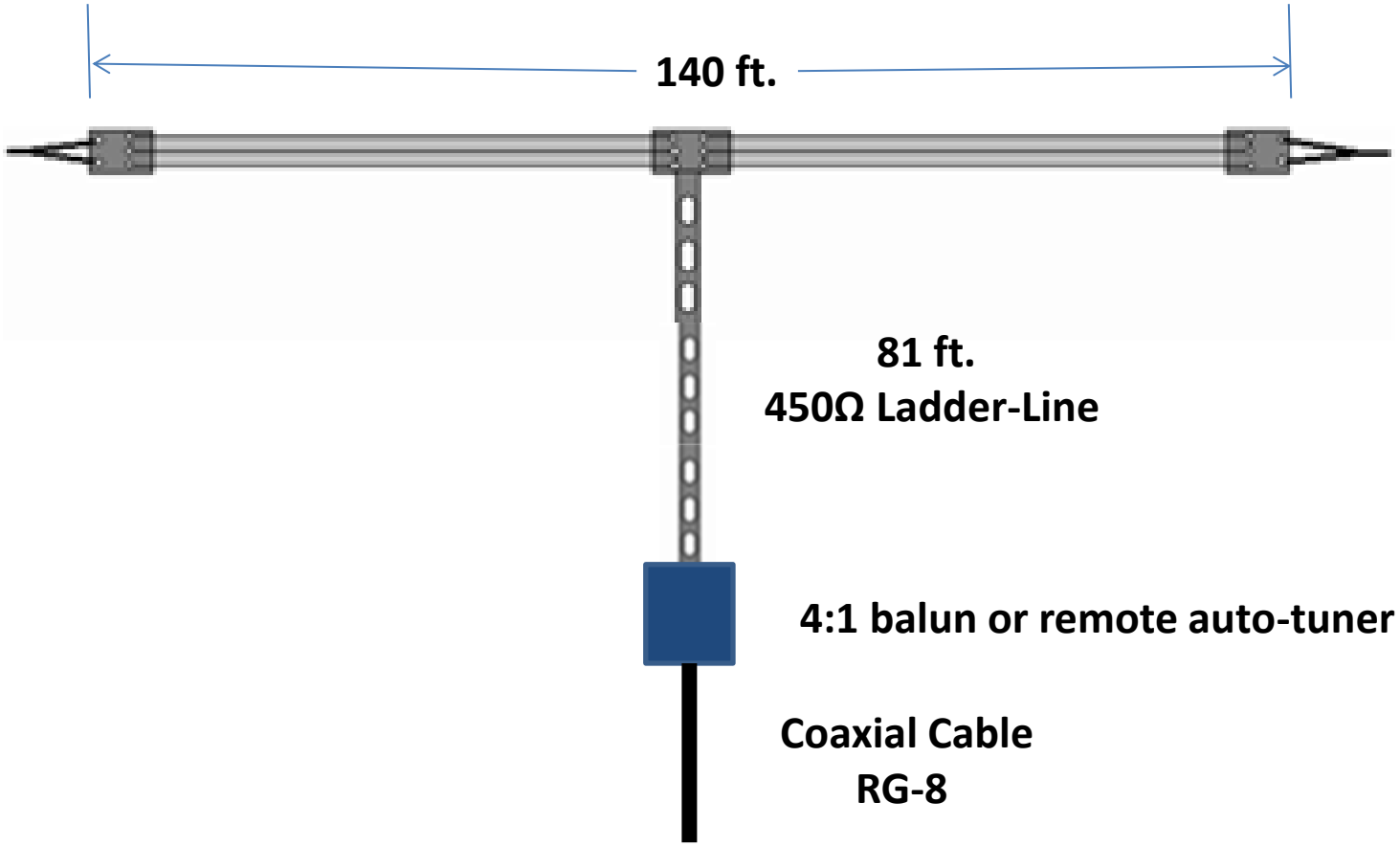
Cobra Ultralite Dipoles



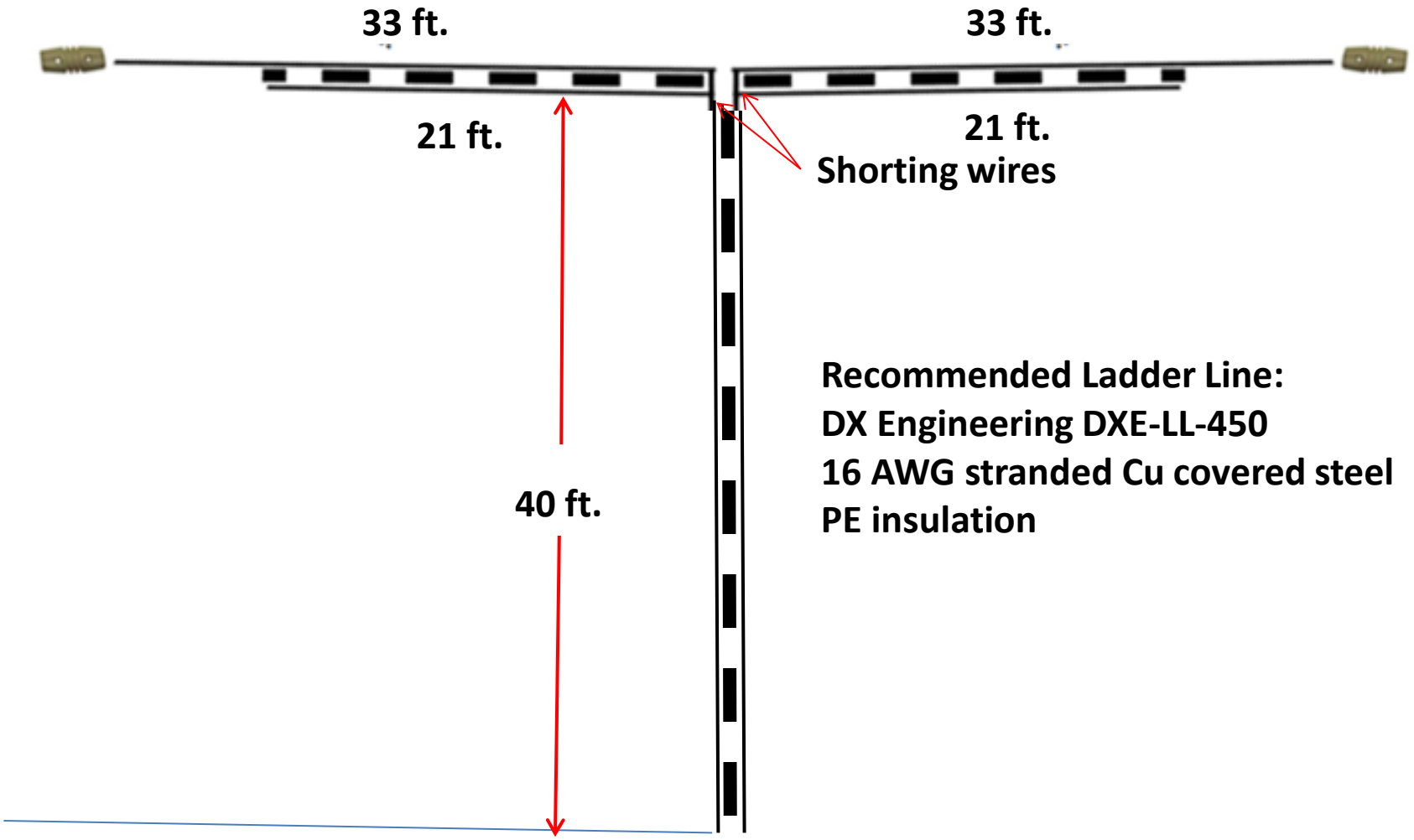
Cobra Ultralite Junior



Cobra Ultralite Senior



Netcom Ladder-Line 66/44



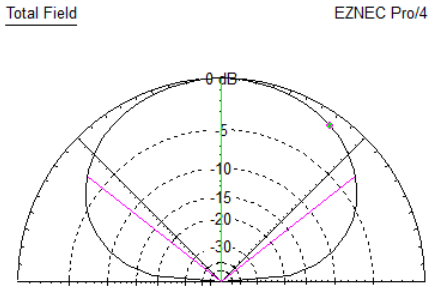
Recommended Ladder Line:
DX Engineering DXE-LL-450
16 AWG stranded Cu covered steel
PE insulation

To 4:1 balun + coax or remote auto-tuner

Final Antenna Selections

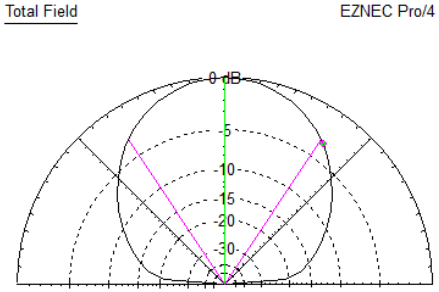
- Horizontal Cobra Junior and Net 66/44 dipole Antennas were designed in EZNEC Pro 4:
  - A single EZNEC command allows height change;
  - The TOA, horizontal beam width and gain are computed and plotted for each entered height.
  - The complex impedance is also available to evaluate the ability of an auto-tuner to find a low SWR solution.
- Two examples are shown on pages 23 and 24.

Azimuth & Elevation – 5.2 MHz Cobra Jr. Horizontal at 21 ft.



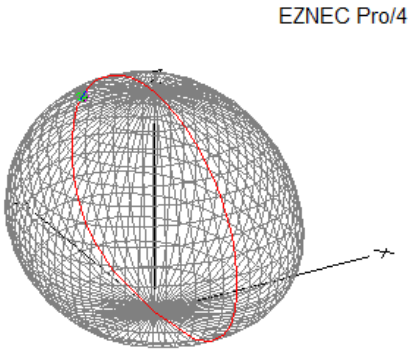
5.202 MHz

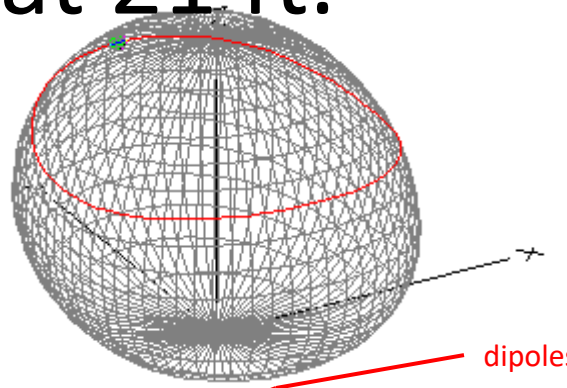
Elevation Plot		Cursor Elev	55.0 deg.
Azimuth Angle	90.0 deg.	Gain	3.42 dBi
Outer Ring	4.56 dBi		-1.13 dBmax
			-1.13 dBmax3D
3D Max Gain	4.56 dBi		
Slice Max Gain	4.56 dBi @ Elev Angle = 90.0 deg.		
Beamwidth	103.6 deg.; -3dB @ 38.2, 141.8 deg.		
Sidelobe Gain	< -100 dBi		
Front/Sidelobe	> 100 dB		



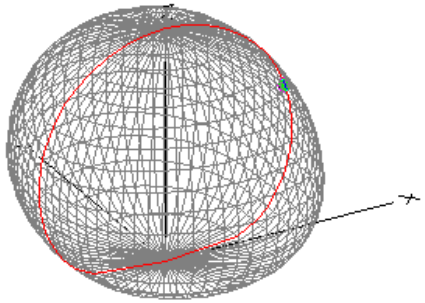
5.202 MHz

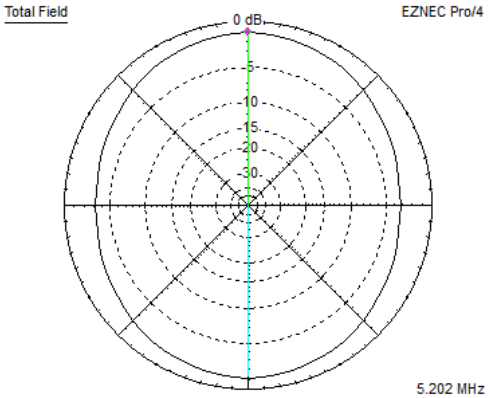
Elevation Plot		Cursor Elev	55.0 deg.
Azimuth Angle	0.0 deg.	Gain	1.32 dBi
Outer Ring	4.56 dBi		-3.24 dBmax
			-3.24 dBmax3D
3D Max Gain	4.56 dBi		
Slice Max Gain	4.56 dBi @ Elev Angle = 90.0 deg.		
Beamwidth	67.2 deg.; -3dB @ 56.4, 123.6 deg.		
Sidelobe Gain	< -100 dBi		
Front/Sidelobe	> 100 dB		





dipoles



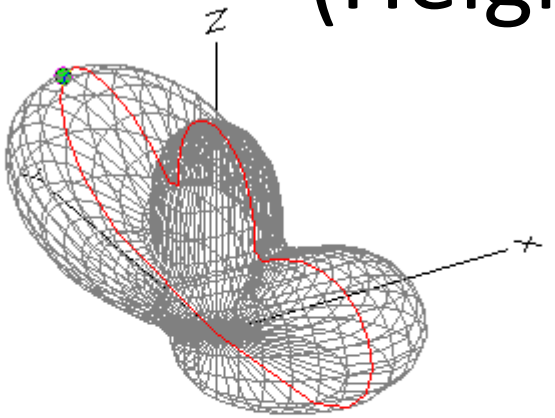


5.202 MHz

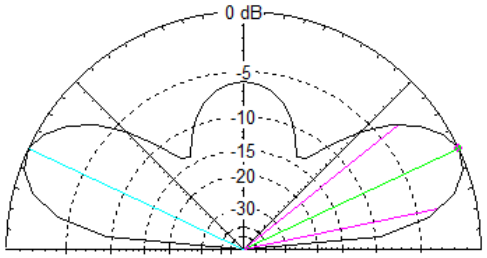
Azimuth Plot		Cursor Az	90.0 deg.
Elevation Angle	55.0 deg.	Gain	3.42 dBi
Outer Ring	4.56 dBi		0.0 dBmax
			-1.13 dBmax3D
3D Max Gain	4.56 dBi		
Slice Max Gain	3.42 dBi @ Az Angle = 90.0 deg.		
Front/Side	2.1 dB		
Beamwidth	?		
Sidelobe Gain	3.42 dBi @ Az Angle = 270.0 deg.		
Front/Sidelobe	0.0 dB		

Frequency 14.2 MHz (Height – 40 ft.)

EZNEC Pro/4



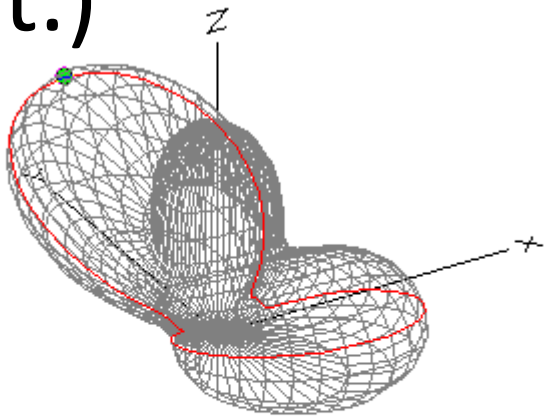
Total Field EZNEC Pro/4



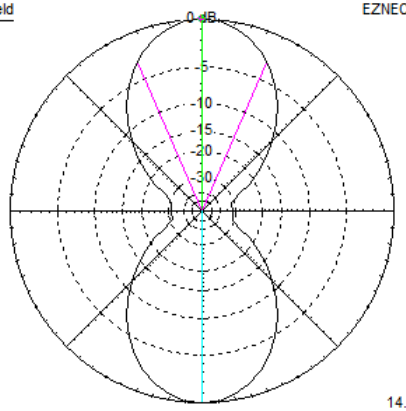
14.2 MHz

Elevation Plot		Cursor Elev	25.0 deg.
Azimuth Angle	90.0 deg.	Gain	9.77 dBi
Outer Ring	9.77 dBi		0.0 dBmax
			0.0 dBmax3D
3D Max Gain	9.77 dBi		
Slice Max Gain	9.77 dBi @ Elev Angle = 25.0 deg.		
Beamwidth	26.9 deg.; -3dB @ 11.8, 38.7 deg.		
Sidelobe Gain	9.77 dBi @ Elev Angle = 155.0 deg.		
Front/Sidelobe	0.0 dB		

EZNEC Pro/4



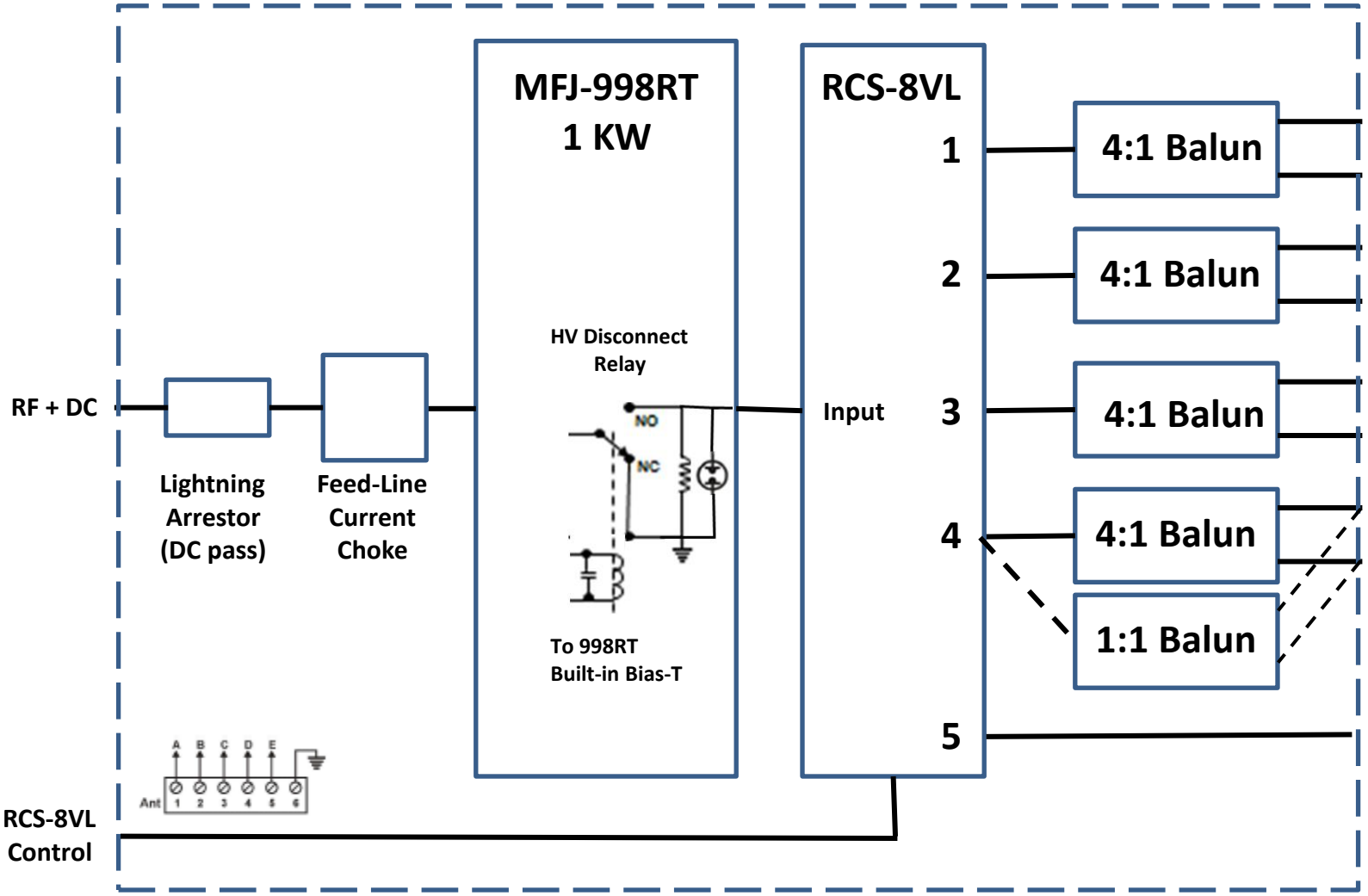
Total Field EZNEC Pro/4



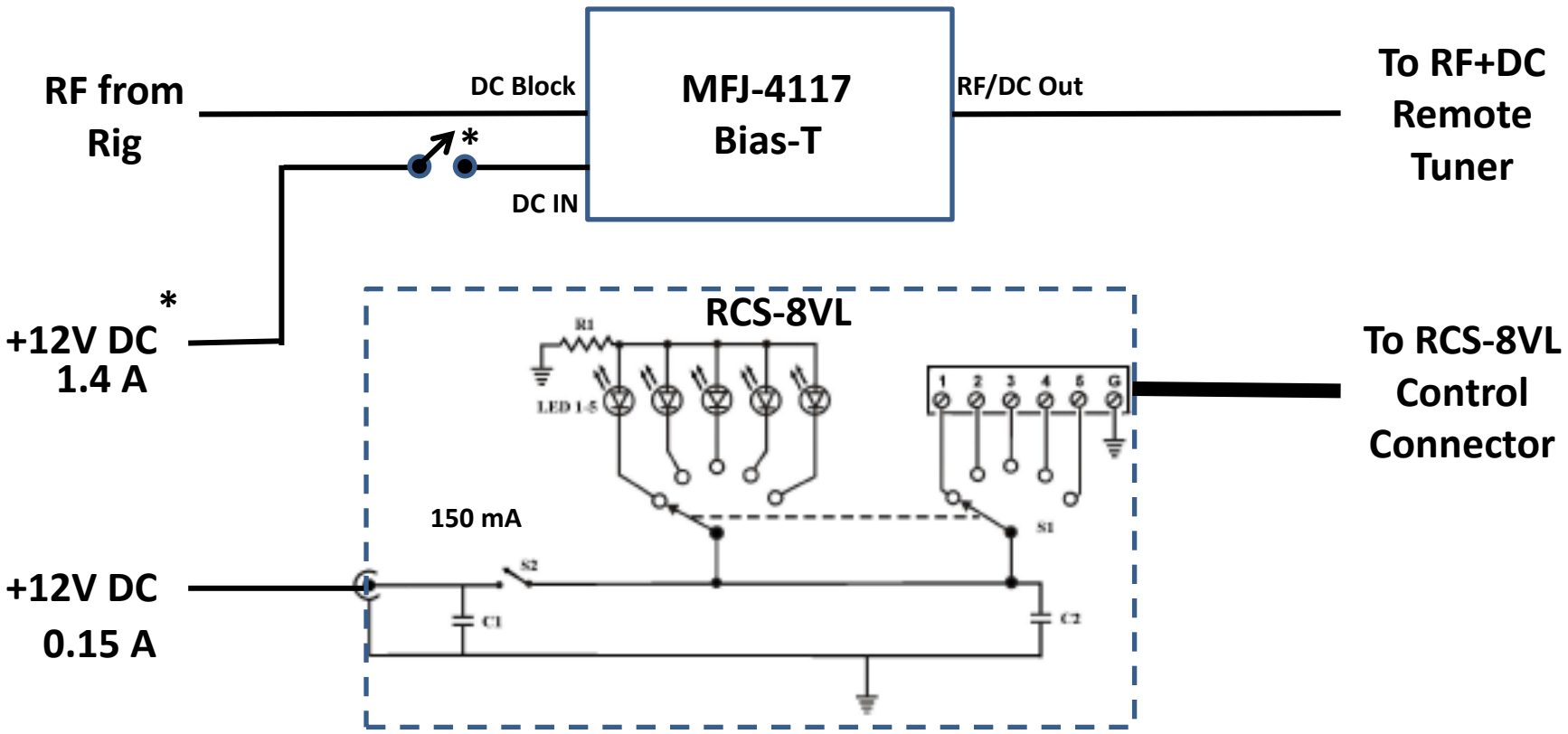
14.2 MHz

Azimuth Plot		Cursor Az	90.0 deg.
Elevation Angle	25.0 deg.	Gain	9.77 dBi
Outer Ring	9.77 dBi		0.0 dBmax
			0.0 dBmax3D
3D Max Gain	9.77 dBi		
Slice Max Gain	9.77 dBi @ Az Angle = 90.0 deg.		
Front/Side	31.43 dB		
Beamwidth	47.4 deg.; -3dB @ 66.3, 113.7 deg.		
Sidelobe Gain	9.77 dBi @ Az Angle = 270.0 deg.		
Front/Sidelobe	0.0 dB		

Remote Tuner with Selectable Balun

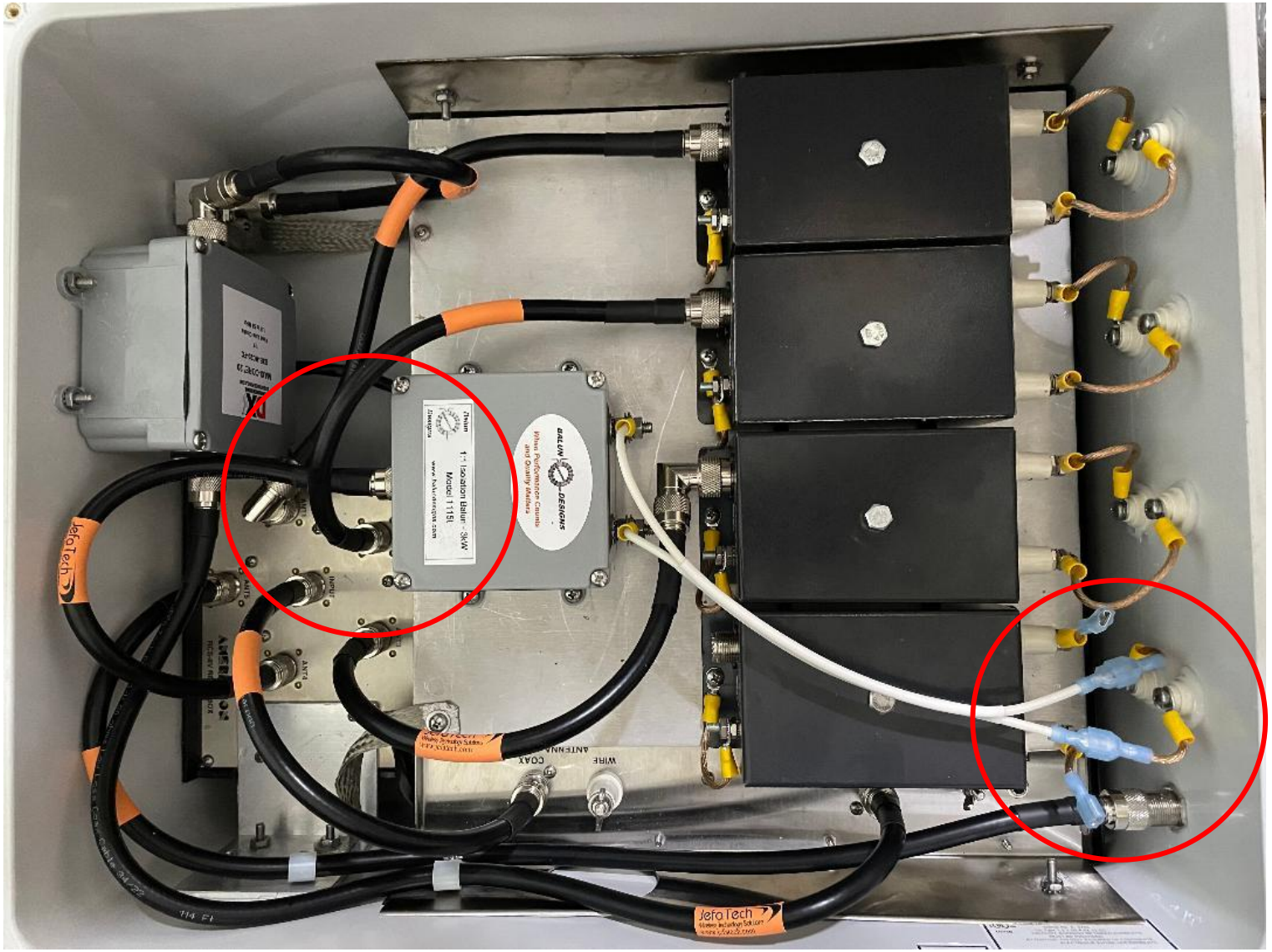


Block Diagram – Station Controls

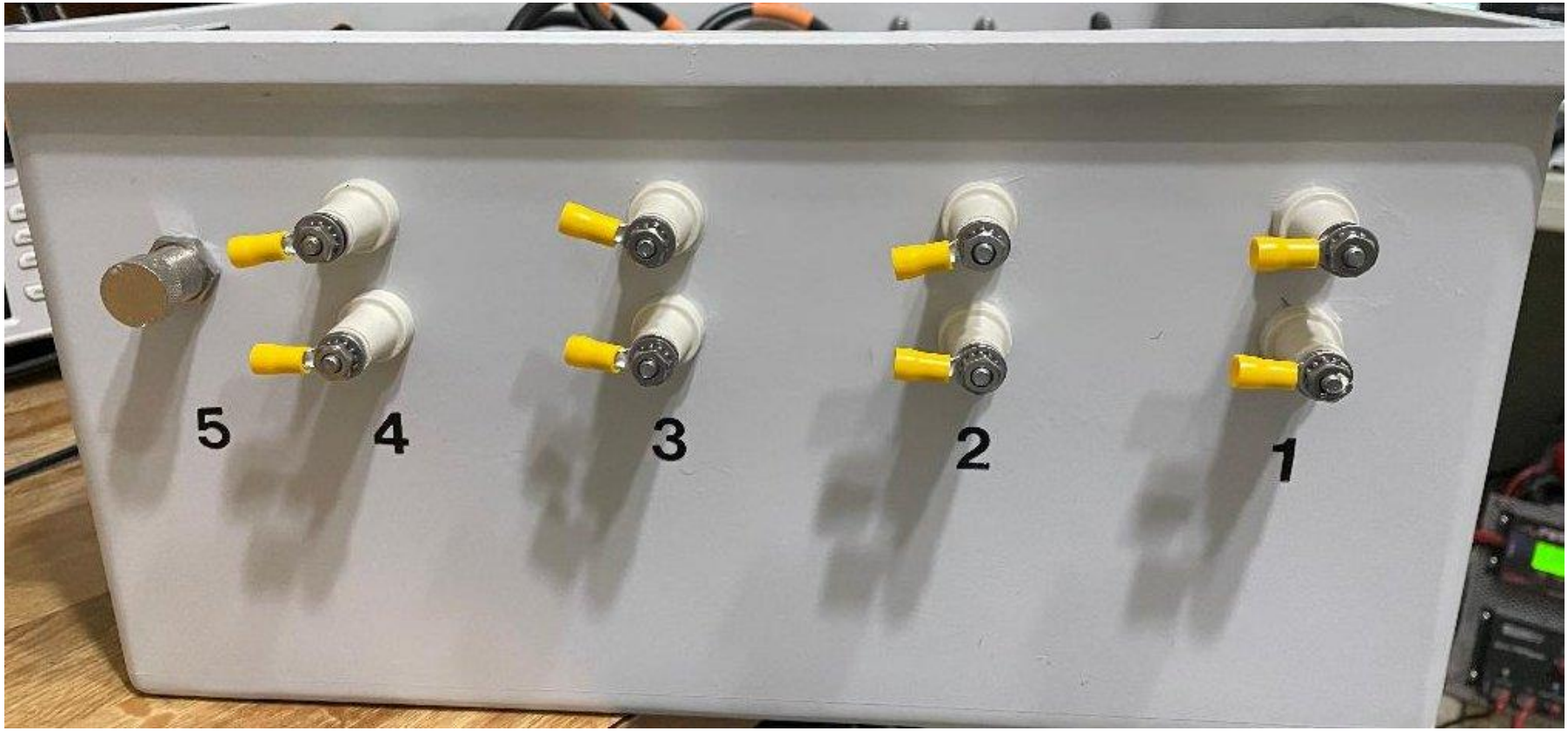


- *Note: A 12 Vdc Adaptor is supplied with RCS-8VL, but an additional 1.4 A is required for the MFJ 998RT, so total required current is 1.55 A DC
- It is important to be able to cycle DC power to the MJF-998RT to restart tuning. Need separate DC sources or switched DC for Bias-T.

Inside View With Selectable Channel 4 Balun Option connected to 1:1 balun



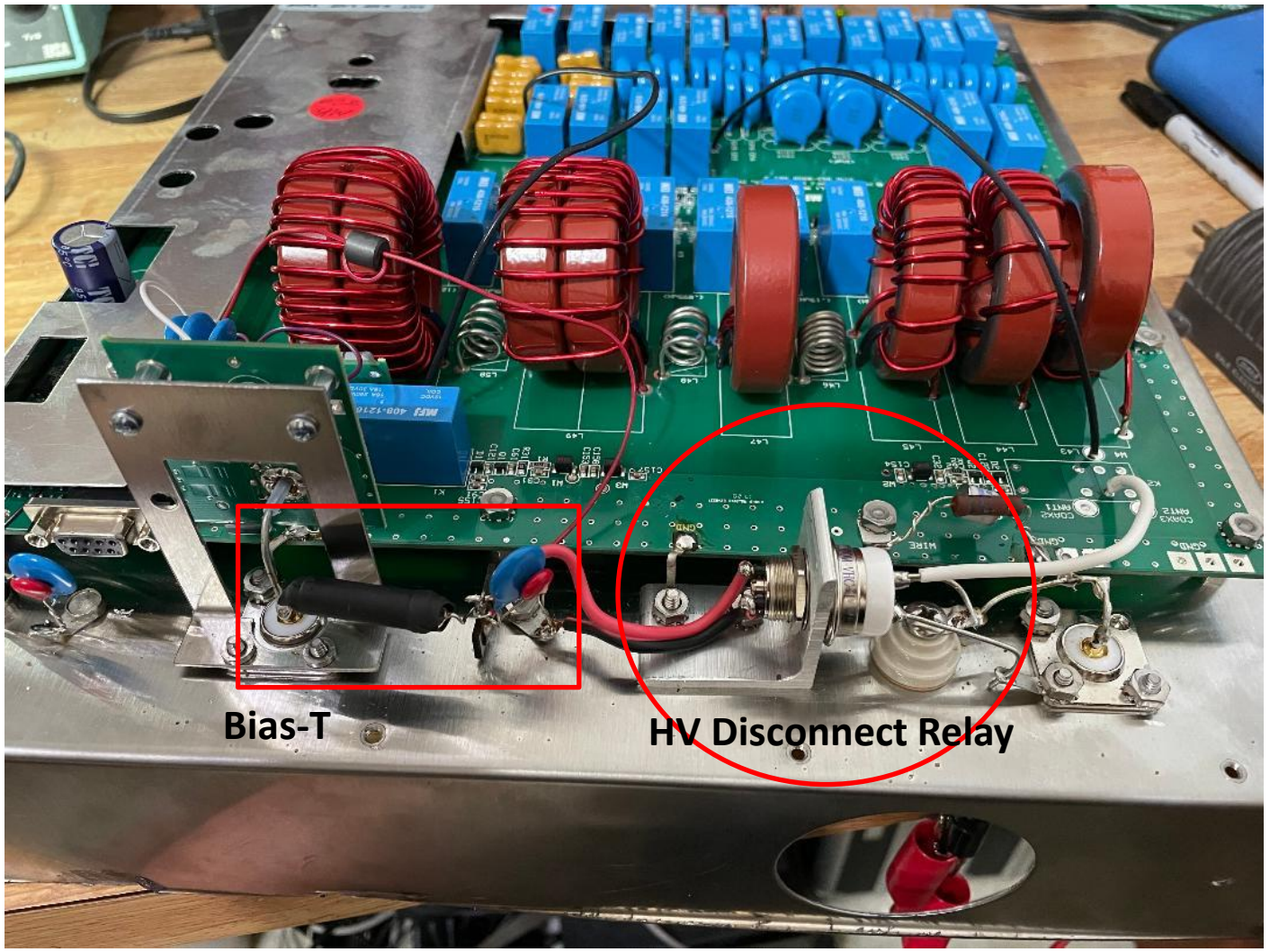
Tuner Box Outputs



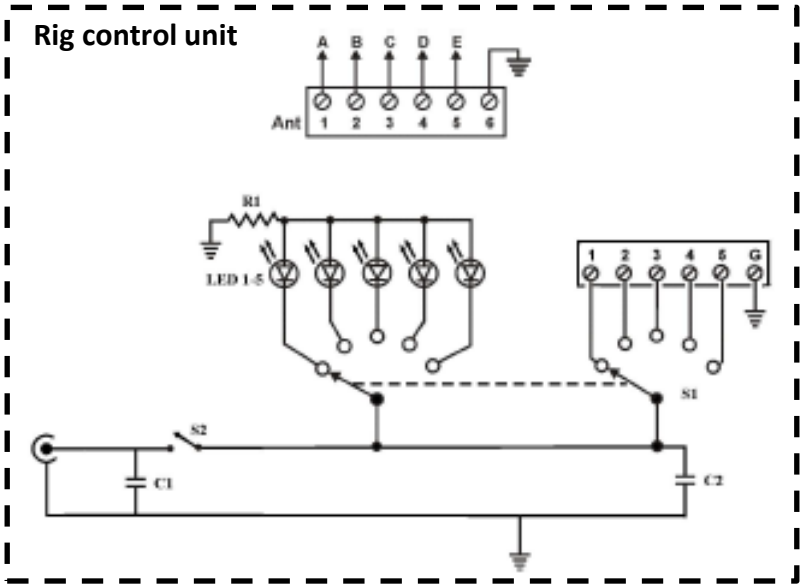
Tuner Box Inputs

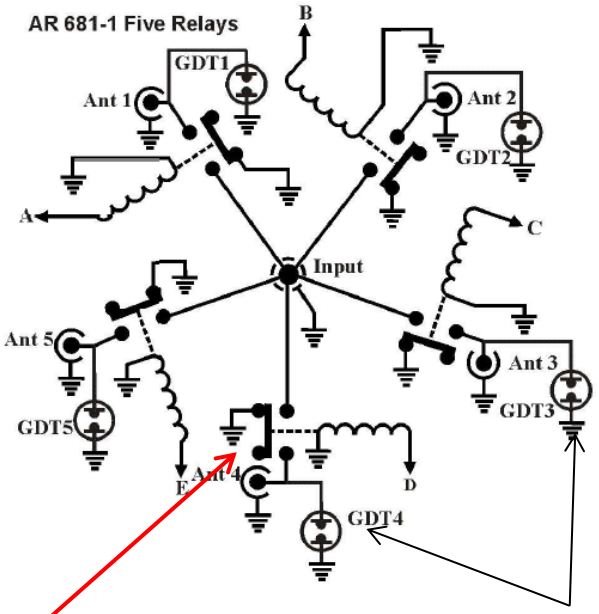


MFJ-998RT Disconnect Relay



Upgraded Ameritron RCS-8VL Antenna Switch

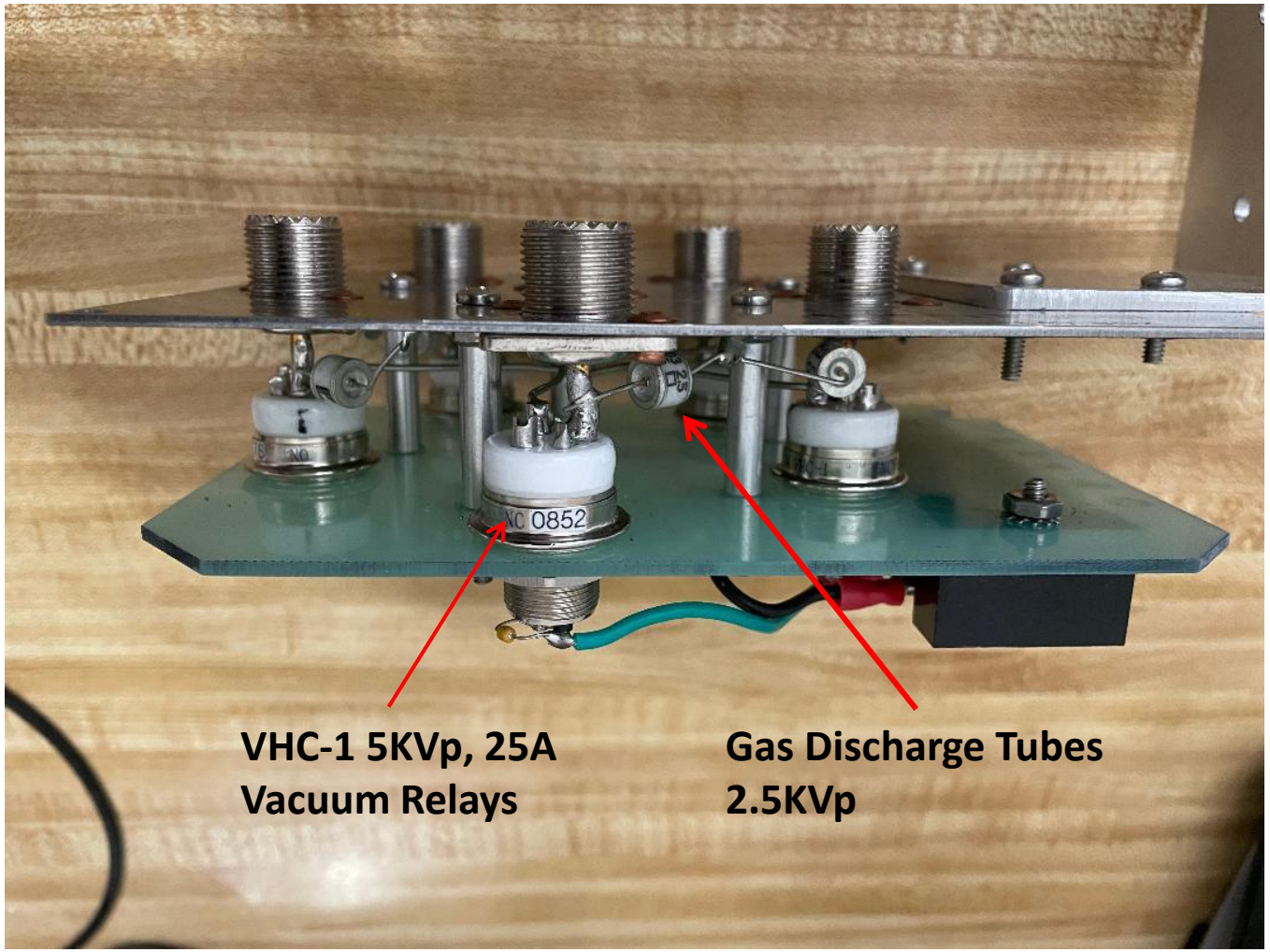




2.5 KV GDT Lightning Arrestors (one per switch)

Note: All relays replaced by VHC-1 Vacuum Relays rated at 5KVp, 25A.

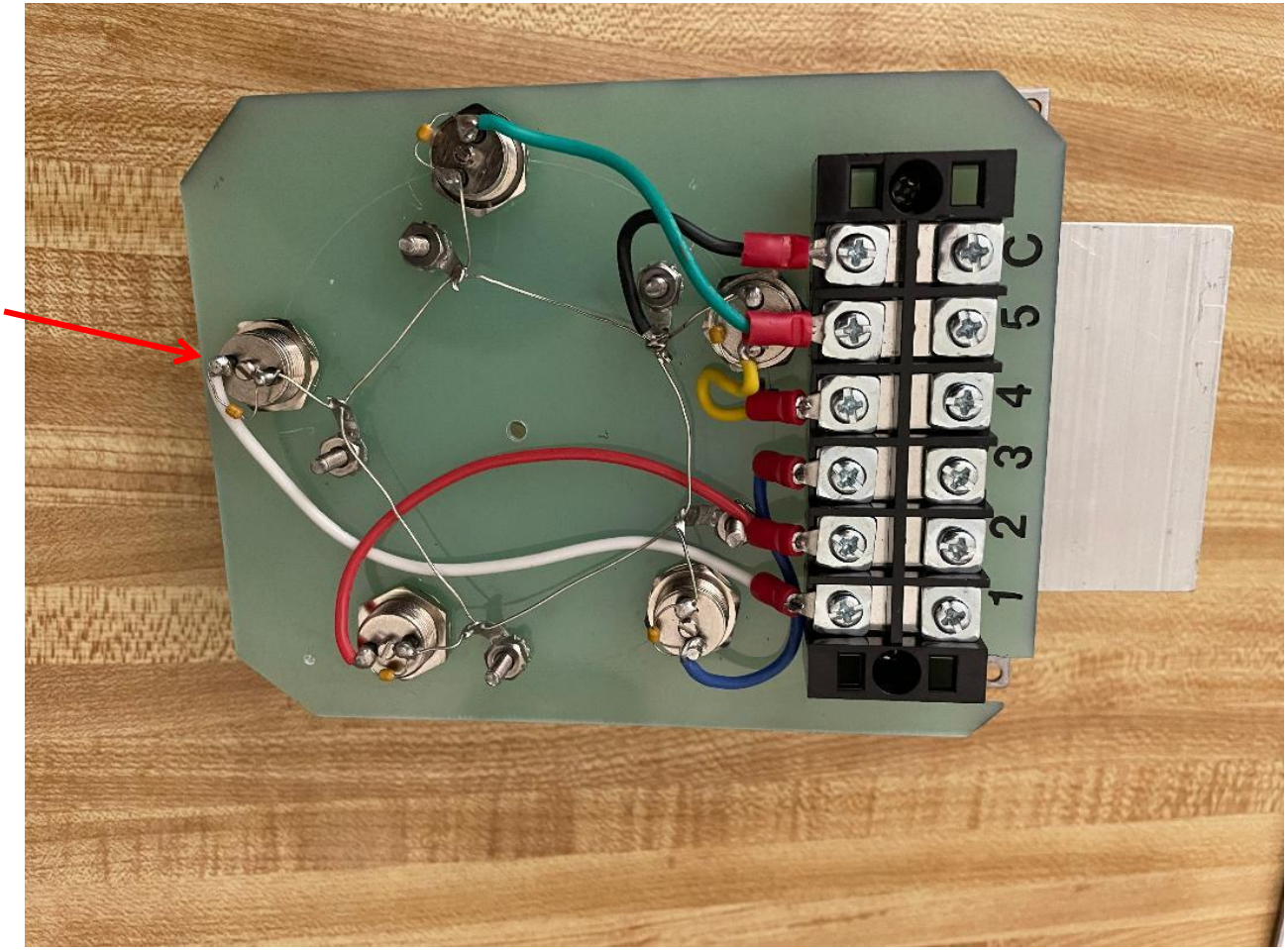
RCS-8VL New HV Relay Assembly



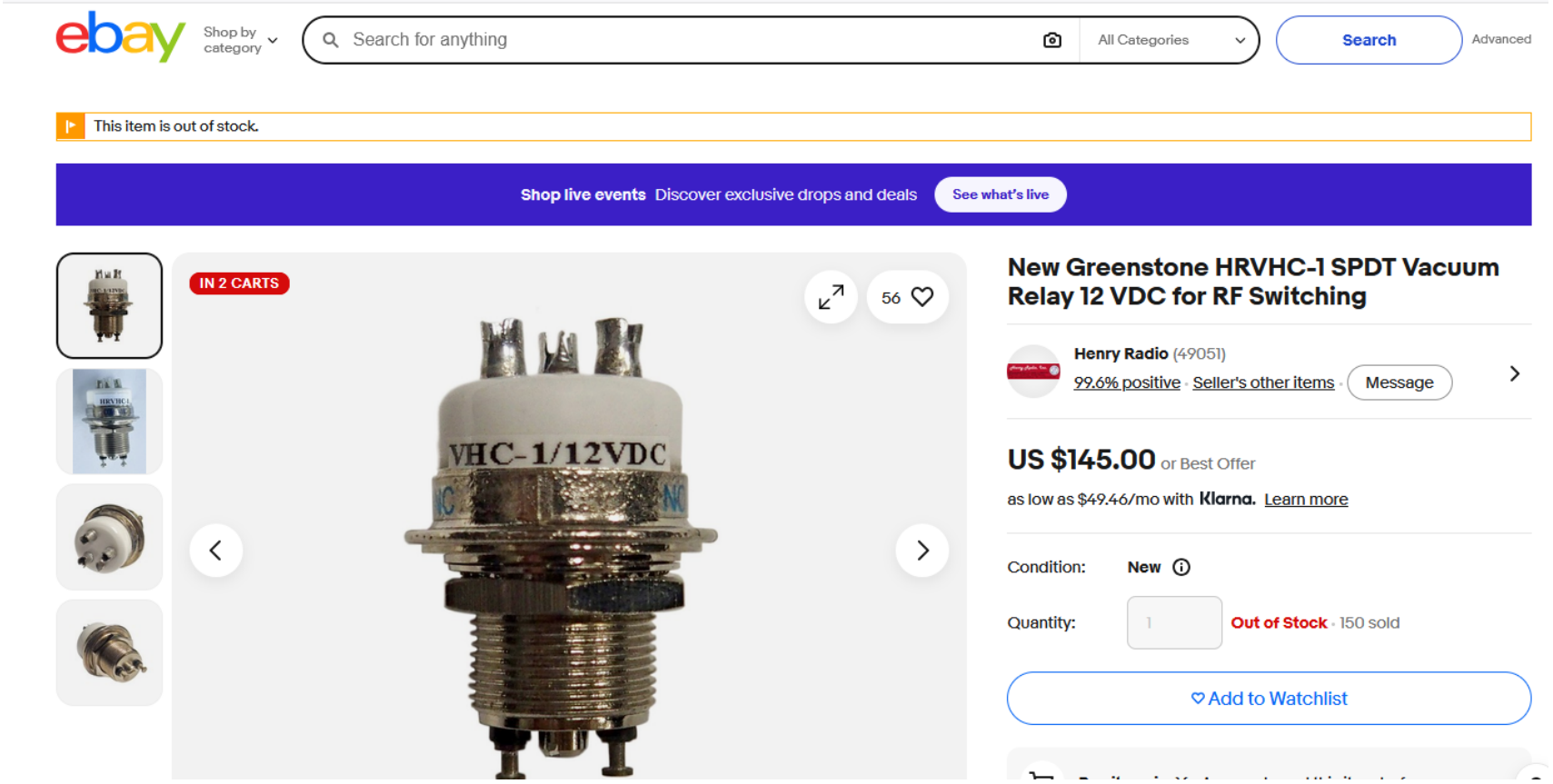
RCS-8V HV Relay Assembly – top view

Access to Relay coils

Unsolder, remove screw, lift out coil

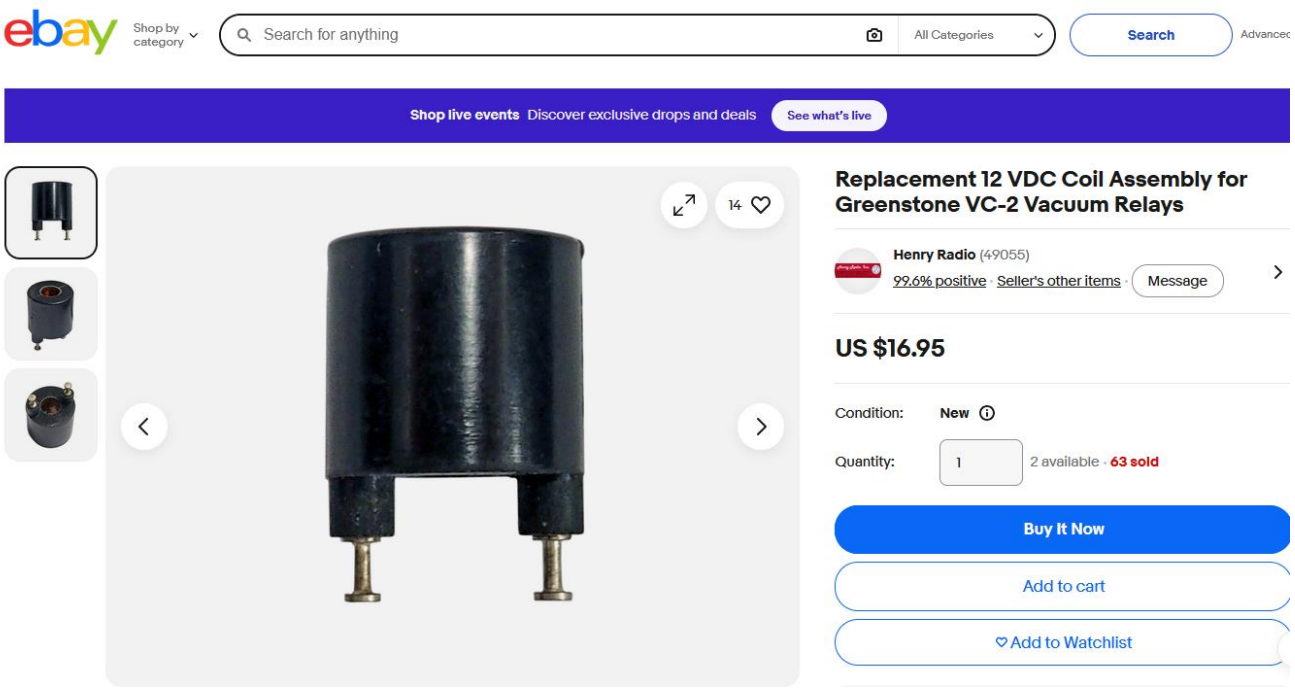


VHC-1 Relay Source

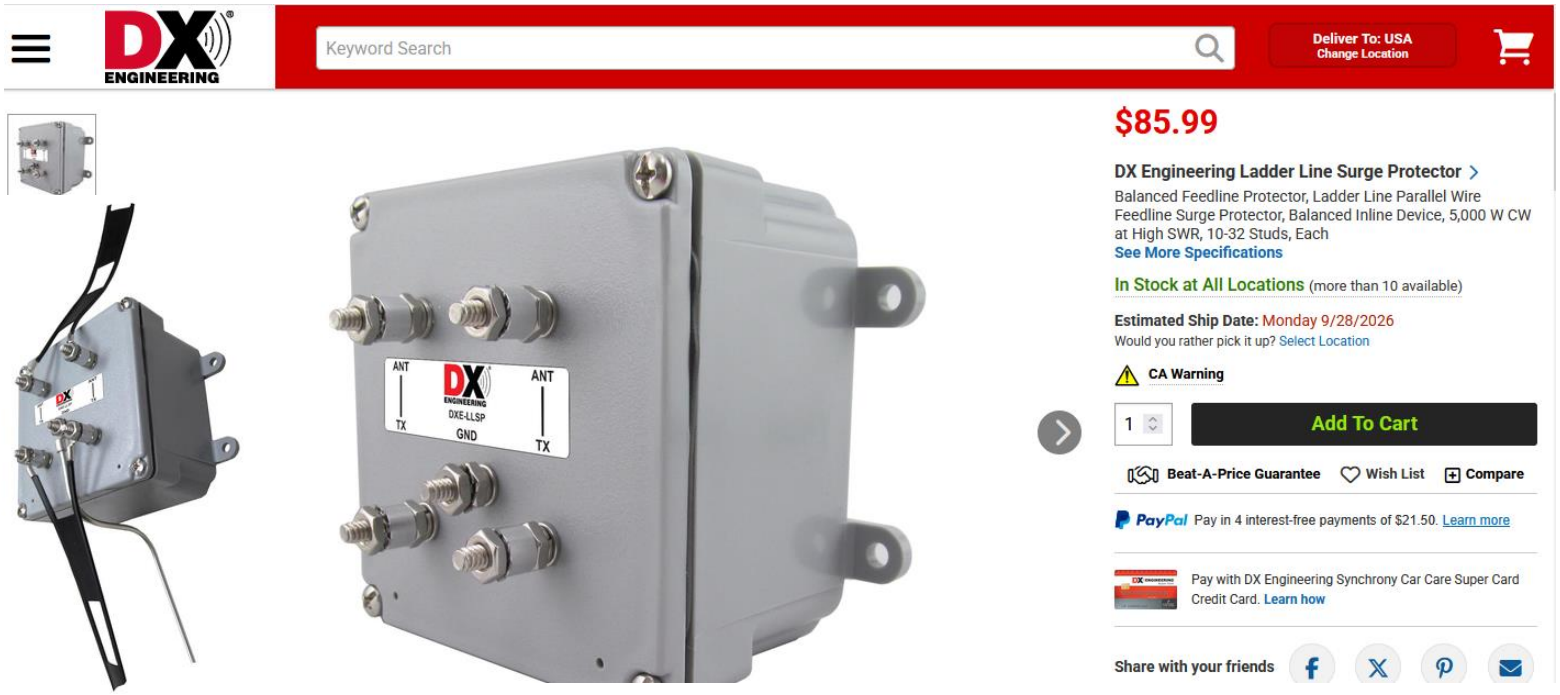


VHC-1 Relay Repair

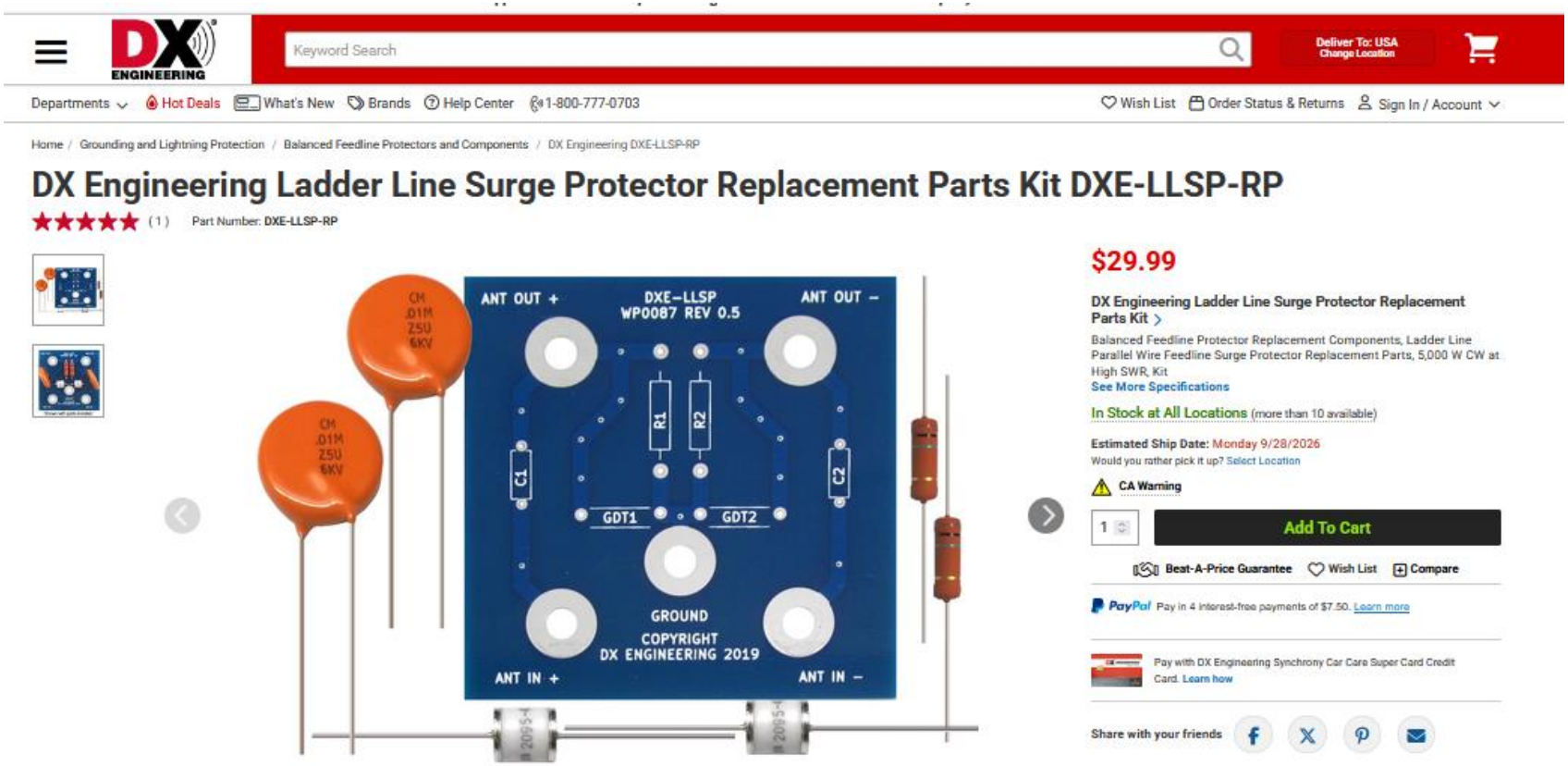
- The relay coil can be easily replaced without replacing the complete relay.



Recommended Additional Lightning Protection For All Antennas



Spares For Ladder-Line Surge Arrestor



Introduction to Major Components

- To follow will be a discussion of each major component in this design.
- This discussion sequence will follow the transmit signal path.

Bias-T

Function – Combines DC (12V) and RF
Transmit power into a single coaxial cable



Specifications

- **Frequency Range:** 1 to 60 MHz DX Engineering
- **Power Handling:** Up to 1,500 watts CW on low SWR feed lines (or 50V / 1A max DC) WiMo +1
- **DC Requirements:** Standard 2.1mm DC plug, 1–50 VDC, 1 amp maximum Universal Radio +1
- **Connectors:** SO-239 RF connectors Universal Radio
- **Physical Dimensions:** 2.75 x 2.5 x 1.25 inches Universal Radio
- **Key Feature:** Built-in On-Off toggle switch to control or interrupt power to remote equipment WiMo +1

Lightning Arrestor

ALPHA-DELTA
ATT3G50UBXL-HP

Surge Protector 2000 Watts to 500 MHz 1.5 Inch Bulkhead UHF Female SO239

HRO Discount Price: $78.95*



- The extended Bulkhead UHF Female SO239 allow unit to be directly mounted
  to the case.
- DC pass allows DC power to be passed through to the bias-T located in the MFJ -998RT auto-tuner.
- The Surge Protector Replacement Arc-Plug® Cartridge is easily replaced if damaged by a lightning strike.

Feed Line Choke

DX Engineering Maxi-Core® DXE-MC20-FC $139.99



Maxi-Core® 20 Current Choke, 1:1, 5 kW, Use Inline on 50 ohm Feedlines, High Common Mode Impedance

The Feed Line Choke is not necessary if you only intend to drive true baluns (balanced to Unbalanced) that then drive balanced antennas. The single-ended auto-tuner is naturally isolated From the balanced antennas by the balun(s).



MFJ-998RT Auto-Tuner

MFJ-998RT, REMOTE AUTO TUNER, 1.5kW, 1.8-30 MHz

$949.95





MFJ-998RT Auto-Tuner

Specifications

- Impedance matching range: 12 to 1600 ohms
- SWR matching range: up to 4:1 for < 50 ohms and up to 32:1 for > 50 ohms
- Minimum power for tuning: 5 watts
- Maximum power while tuning: 100 watts with foldback, 20 watts without foldback
- RF power limit: 1500 watts SSB/CW
- Frequency range: 1.8 to 30 MHz continuous coverage
- Frequency counter accuracy: ±1 kHz across HF bands
- Capacitance range: 0 to 3926 pF nominal (256 values) on input side 0 to 976 pF nominal (64 values) on output side
- Inductance range: 0 to 24.28 µH nominal (256 values)
- Relay rating: 16 amp 1000 volts
- Relay electrical life: 100,000 operations
- Relay mechanical life: 10 million operations
- Memory endurance: 1 million erase/write cycles
- Memory data retention: > 200 years
- Power requirements: 12 - 15 volts DC, 2.1 × 5.5 mm coaxial plug, center pin positive
- Current consumption: 1.4 amps or less
- Dimensions (approx.): 13 ¾ × 6 ¼ × 18 in. (350 × 159 × 457 mm) (width/height/depth) including connectors
- Weight (approx.): 9 lb. 8 oz. (4.3 kg)
- Operating Temperature -40 to +160 degrees F

There is a built-in Bias-T that allow remote operation through one coaxial Cable. A MFJ- 4117 bias-T is supplied with the MFJ-998RT

Ameritron RCS-8V Antenna Switch

AMERITRON RCS-8V

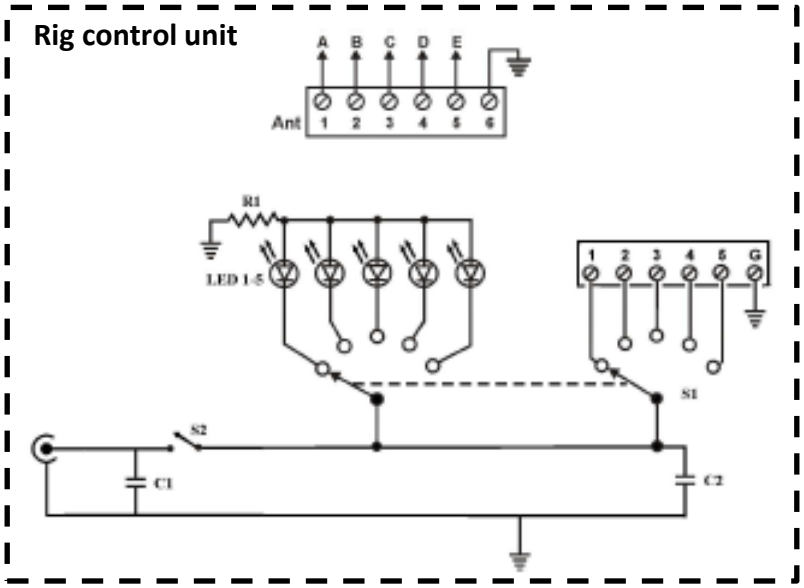
Remote Coaxial Antenna Switch 5 Position 110 Volts

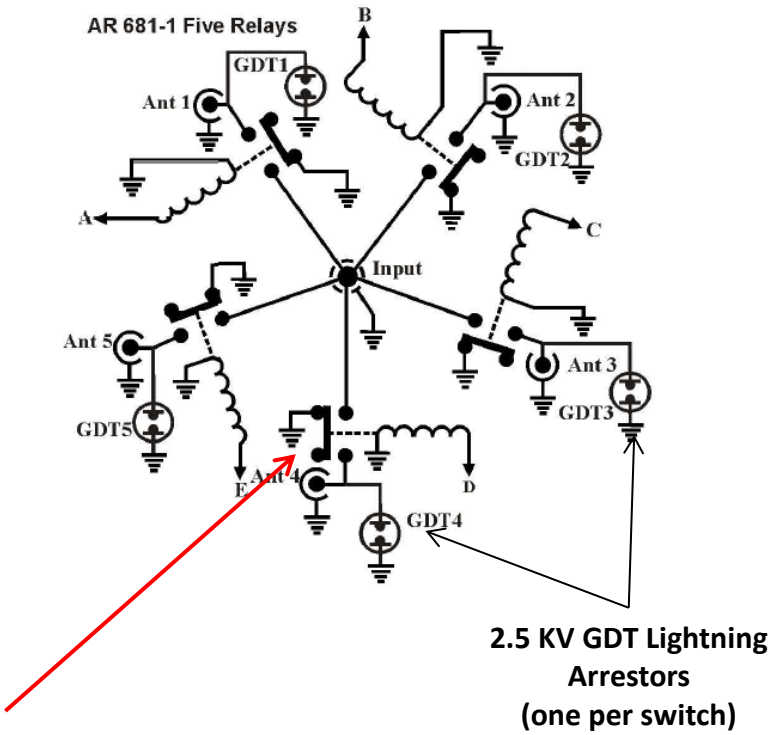
HRO Discount Price: $274.95*



The Ameritron RCS-8V is a remote controlled coaxial RF switch that will operate with negligible loss, radiation and VSWR at all frequencies up to 250 MHz. Only a slight compromise in VSWR occurs at 450 MHz. Power rating is four kilowatts PEP below 30 MHz and one kilowatt PEP at 150 MHz.

Upgraded Ameritron RCS-8VL Antenna Switch





Note: All relays replaced by VHC-1 Vacuum Relays rated at 5KVp, 25A.

RCS-8VL Gas Discharge Tubes

CG32.5L - CG3 Series

Series: CG3



Property (Mouseover for details)	Value
MIN Sparkover Voltage (V)	2000
Nom DC Sparkover (V)	2500
Max DC Spark over Voltage	3000
Max Impulse Sparkover Voltage @ 100V/us (V)	3750
Capacitance (Co) (pF)	1
Impulse Discharge Current @8/20µs (kA)	10
Electrodes	2

DXE-IS-RCT Cable Lightning Protection

This unit is install at the house entrance. The RCS-8VL has its own set of lightning arrestors.



$169.99

DX Engineering Rotator Control Line Protectors >

Control Line Surge Protector, 8-Terminal Shunt Type, 2 A, 60 Vac or 70 vdc max., Each

Overview

Brand:	DX Engineering
Manufacturer's Part Number:	DXE-IS-RCT
Part Type:	DC Shunt Control Line Protectors
Product Line:	DX Engineering Rotator Control Line Protectors
DXE Part Number:	DXE-IS-RCT
Protector Line Voltage:	82 Vac
Quantity:	Sold individually.
Notes:	AC or DC voltage spikes exceeding 82 Volts (in either polarity) will be shunted to ground.

4:1 Baluns - MFJ

MFJ MFJ-912

W9INN Remote Balun Box - 1.8 to 30 MHz - 1500 Watts - for Antennas Fed with Balanced Feedline

HRO Discount Price: $89.95*



Giant 2-core 4:1 balun wound with Teflon wire connects to high voltage ceramic feedthru insulators.
Handles full legal limit with ease. SO-239 connector.



1:1 Balun

BALUN DESIGNS

Model 1115 – 1:1 Balun, 1.5–54 MHz, 3kW $78.95



Feed-thru Insulators (Two per balun)

MGS MAX-GAIN SYSTEMS, INC.

$7.95

High Voltage, Ceramic Feed Through Insulator

Quantity Discounts Available:

(1-9 QTY) Price: **$7.95** each

(10-24 QTY) Price: **$7.16** each **Save 10%**

(25-99 QTY) Price: **$6.36** each **Save 20%**

(100+ QTY) Price: **$5.57** each **Save 30%**





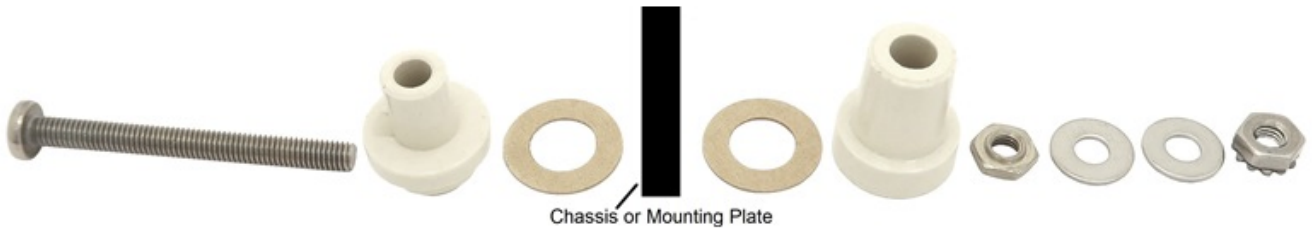
FT-IN-01 Feed-Thru

High Voltage, Ceramic Feed Through Insulator

Manufacturer's Voltage Spec:	Unknown
Max-Gain Systems Bench Testing Results	14 KV (before any leakage detected)
Screw length:	2.095 in
Screw Diameter:	.183 in
Screw Material:	Stainless Steel
Ceramic Outer Diameter	.755 in
Ceramic Inner Diameter (Mounting Hole)	.384 in
Screw Thread	10-32

Installation and use

These Feed Through Insulators will fit on and any plate or chassis up to about .25 inches thick. Perfect for all applications. Parts shown below in order of assembly. The hole diameter that needs to be drilled needs to measure at least 0.383 inches.



Packing Gland for Rotor Cable

Lapp Skintop Cord Grip - PG-11 - .157" - .394" OD-Gray



SKU: S1111

Gray | Thread: PG-11 | Cable Outer Diameter Range: .157"- .394" | Liquid Tight |

Quantity Discount Pricing				
1-9	10-24	25-49	50-99	100+
$2.89	$2.83	$2.65	$2.55	$2.49

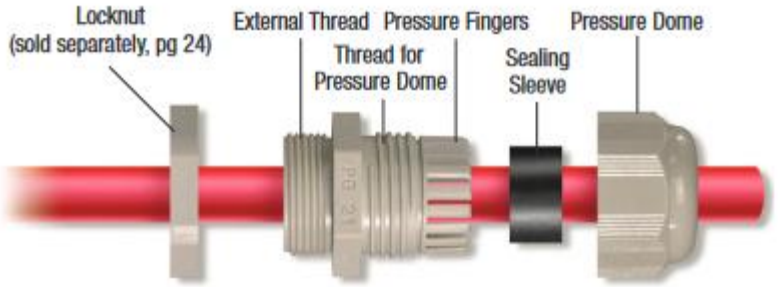
Quantity Available: 0

Qty

1 **Add to Cart** **Call for Availability**

Add to Shopping List Add to Compare

This eliminates the expensive mating waterproof connectors used in a previous design.



Enclosure

TEquipment

(877) 571-7901
Sales@TEquipment.NET



Cover

Lift-Off Cover

H x W x D (Inches) **Inside Dimensions**

20 x 16 x 8.13

N4X Wallmount Encl - 20.00 x 16.00 x 8.13 - Fiberglass

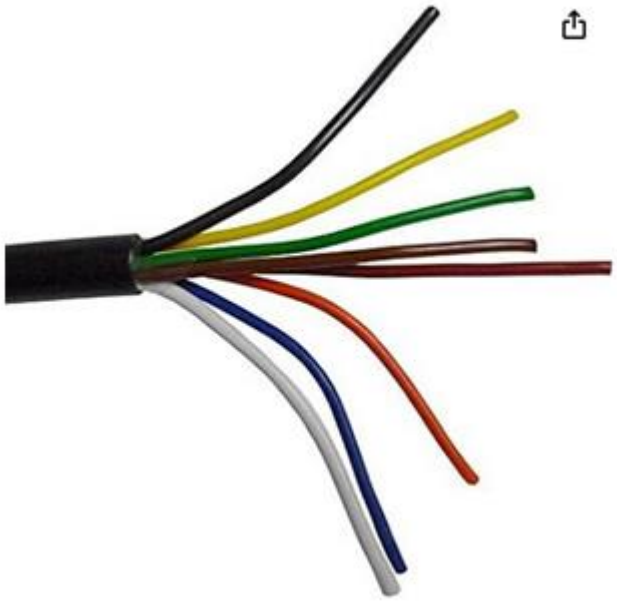
SKU: **Hammond PJ20168**

Price as configured

$346.90

Cables: Rotor (For RCS-8VL)

JSC 5972-8 Conductor Rotor Cable, 16 AWG x 2 + 18



About this item

- Heavy Duty Rotor Wire
- True Copper (Not Copper Clad)
- 6 Wires 18 Gauge, 2 Wires 16 Guage
- Color Varies: Black or Gray

HAM RADIO OUTLET
hamradio.com

Order Num: WT-1345672	UTC Order Date: Dec 13, 2023

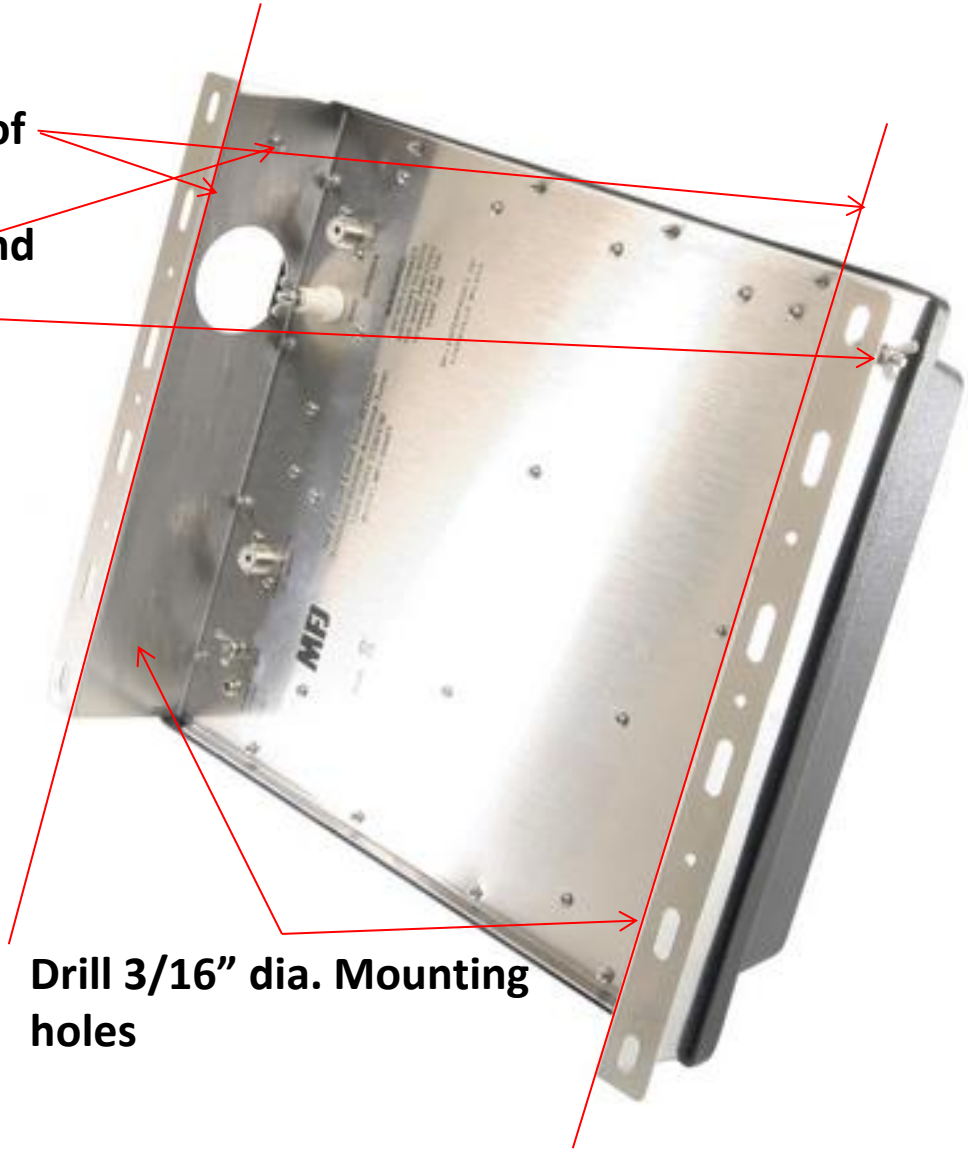
BILLING INFORMATION	RECIPIENT INFORMATION
LEWIS THOMPSON (w5ifq) 5221 MCDADE DR AUSTIN, TX 78735 Email: w5ifq@att.net Phone: 5125879944	LEWIS THOMPSON 5221 MCDADE DR AUSTIN, TX 78735 -Comments-

ORDER ITEMS

LINE	ITEM	PRICE	QTY	SUB
1	**JSC** HEAVY DUTY #5972	$1.49	70	$104.30
				Sub Total: $104.30
				Shipping: $0.00
				Ship Method: Standard
				Tax (8.25%): $8.60
				Paid By Visa | **Total: $112.90**

MFJ-998RT Mechanical Modifications

1. Cut ears off of mounting frame
2. Remove ground Lugs

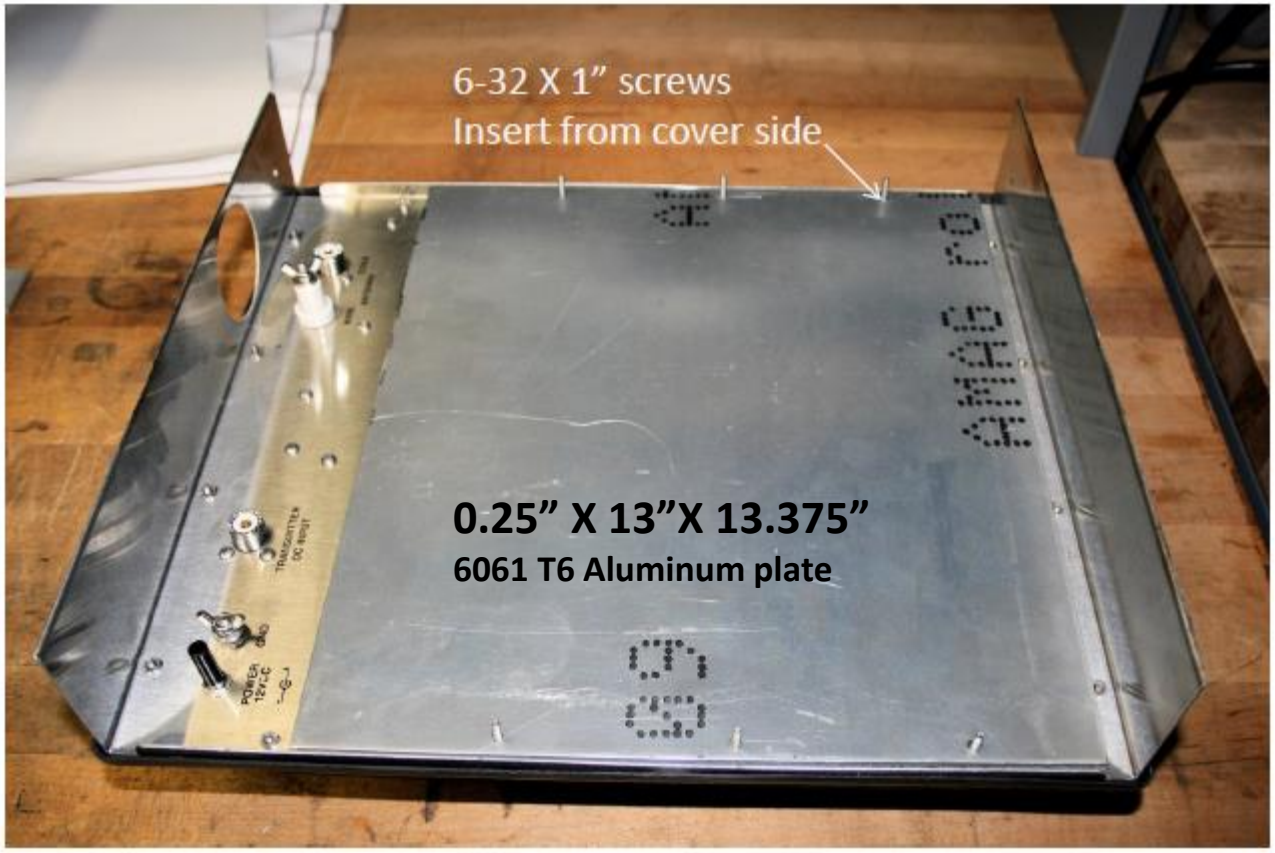


Drill 3/16” dia. Mounting holes

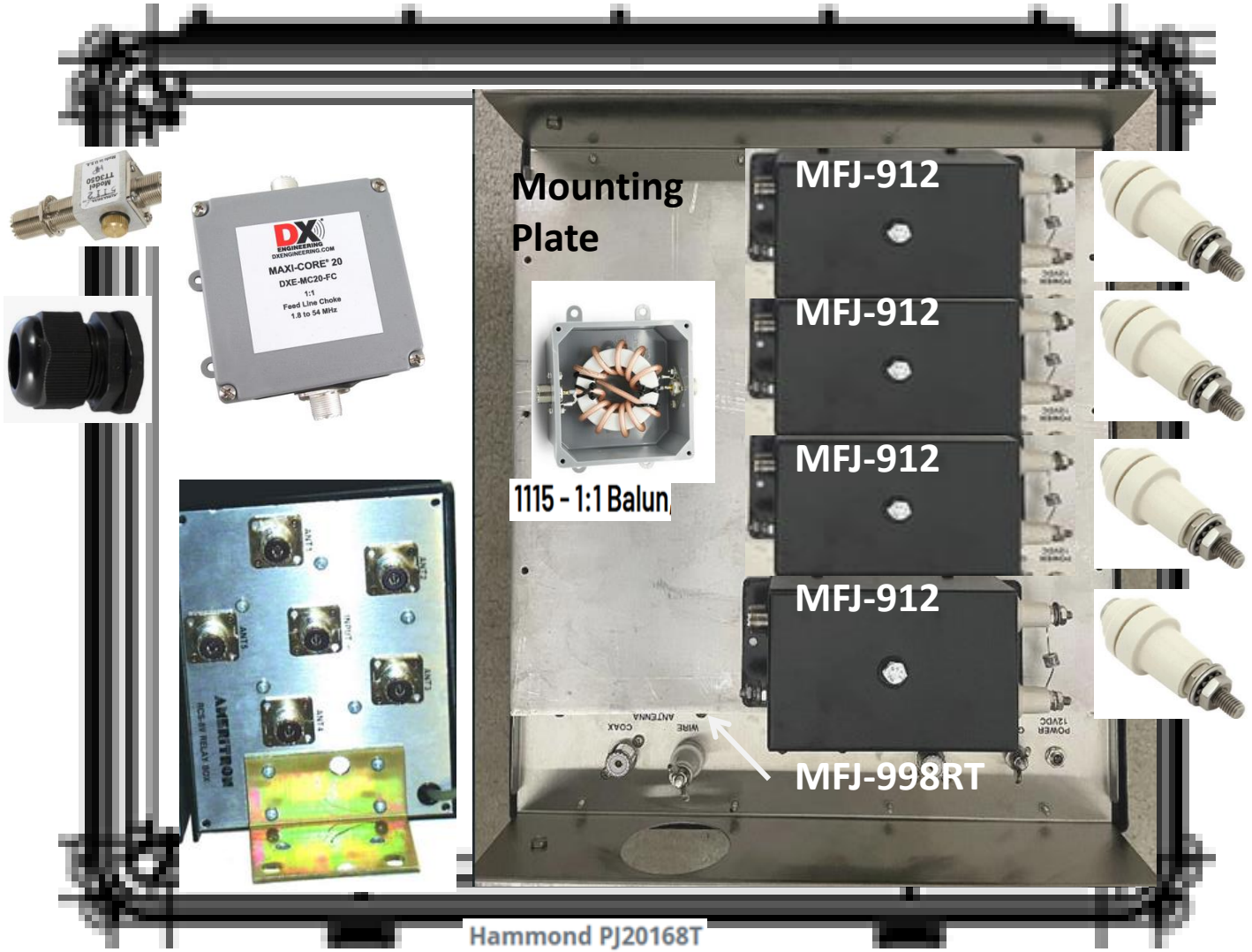
RCS-8VL Mechanical Modifications



Misc. Hardware – MFJ-998RT Balun Mounting Plate



Layout of Components in Enclosure



Two Feed-Throughs Per balun

Components Mounted in Box



Sealant Used on all NMEA Case Penetrations

$8.48

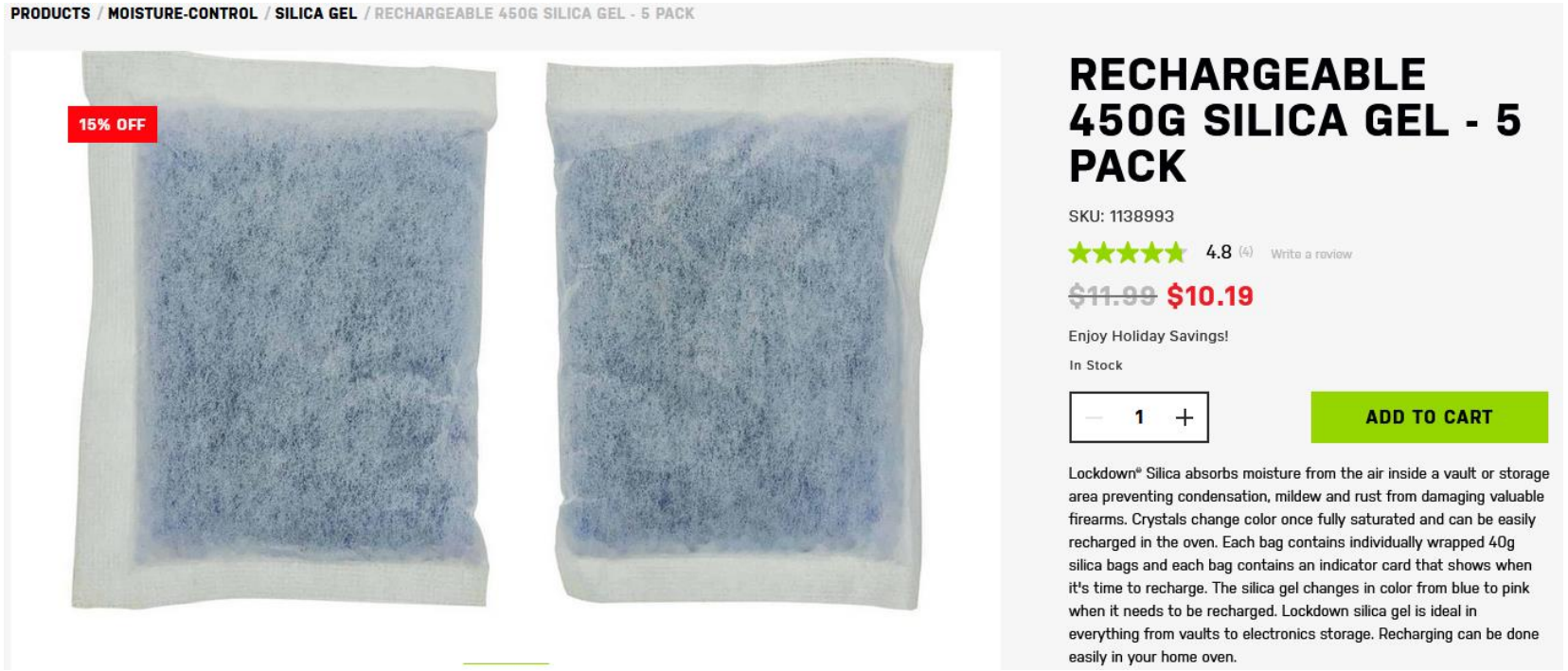
GE

2.8 oz. Supreme Silicone Clear Kitchen and Bath Caulk

- 100% waterproof and 30-minute water-ready
- Mold resistant with lifetime mold-free product protection
- Excellent adhesion and flexibility with 50% joint movement



Humidity Control In NMEA Enclosure



Anti-Seize Conducting Paste

Jet-Lube SS-30 Pure Copper Anti-Seize 12502

★★★★★ (7) Part Number: **JTL-12502**





$42.99

Jet-Lube SS-30 Pure Copper Anti-Seize >

Anti-Seize, Jet-Lube SS-30 Pure Copper, Anti-Oxidant Corrosion Inhibiting, 0.50 lb., Brush Applicator, Each
See More Specifications

In Stock (more than 10 available)

Estimated Ship Date: Today
Would you rather pick it up? Select Location

Restricted

1 Add To Cart

Wish List Compare

Cost - $3888.63

item	MANUFACTURER	SOURCE	MODEL	COST/EACH	QUAN	ORDERED?	DELIVERED?	Net Total	Sales Tax	TOTAL
Auto-Tuner, MFJ	MFJ	MFJ	MFJ 998RT	949.95	1	YES	YES	$949.95	$78.37	$1,028.32
Antenna Switch	Ameritron	HRO	RCS-8V	274.95	1	Yes	YES	$274.95	$22.68	$297.63
HV Upgrade to RACS-8V	Thompson Engr	Thompson Engr.		717.54	1	YES	YES	$717.54	$59.20	$776.74
4:1 Balun	MFJ	MFJ	MFJ-912	89.95	2	YES	YES	$179.90	$14.84	$194.74
4:1 Balun USED	MFJ	MFJ	MFJ-912 USED	50.00	2	YES	YES	$100.00	$0.00	$100.00
1:1 Balun	Balun Designs	Balun Designs	1115	91.70	1	YES	YES	$91.70	$7.57	$99.27
Feed Line Current Choke	DX ENGINEERING	DX Engineering	DXE-MC20-FC	139.99	1	YES	YES	$139.99	$11.55	$151.54
Lightning Arrestor, coaxial	ALPHA DELTA	HRO	ATT3G50UBXL-HP	78.95	1	YES	YES	$78.95	$6.51	$85.46
Lightning Arrestor for control cable	DX ENGINEERING	DX-ENGINEERING	DXE-IS-RCT	169.99	1	YES	YES	$169.99	$14.02	$184.01
RCS-8V Control Cable	JSC	HRO	#5972	1.49	70	YES	YES	$104.30	$8.60	$112.90
Coaxial cable jumpers, 1 ft.	JEFA Tech	Amazon	RG-213 jumpers	15.95	7	YES	YES	$111.65	$9.21	$120.86
Coaxial cable jumpers, 2 ft.	JEFA Tech	Amazon	RG-213 jumpers	17.95	2	YES	YES	$35.90	$2.96	$38.86
PL259 Right Angle Adapter	RFAdapter	Amazon		6.00	3	YES	YES	$18.00	$1.49	$19.49
Feed-thru Insulator	unknown	MGS	FT-IN-01	7.16	10	YES	YES	$71.60	$5.91	$77.51
Packing Gland	Lap Skintop	Lap Skintop	PG-11	0.00	1	in stock	YES	$0.00	$0.00	$0.00
NMEA Case	HAMMOND	TE EQUIPMENT	PJ20168	411.41	1	YES	YES	$411.41	$33.94	$445.35
Al Plate, 6061-T6, 0.25 x 13" x 13.375"	WEST BROOK METALS	WEST BROOK METALS	THOMPSON ENGR.	81.96	1	yes	YES	$81.96	$6.76	$88.72
Al Angle Bracket, 3", .125 wall	Online metals	Online Metals	THOMPSON ENGR.	22.80	1	YES	YES	$22.80	$1.88	$24.68
Sillica Gel Packs	Lockdown	Lockdown	450g 5 pack	15.34	1	yes	YES	$15.34	$1.27	$16.61
Round spacer #6, 5/1" OD, 3/8"	Chenango Supply	Amazon	310606RSA	5.91	1	yes	YES	$5.91	$0.49	$6.40
Split Ring Lock Washers #10	Chenango Supply	Amazon	SS 18-8	8.81	1	yes	YES	$8.81	$0.73	$9.54
Misc wire and crimp ring terminals				5.00	1	IN STOCK	YES	$5.00	$0.00	$5.00
Misc hardware				5.00	1	IN STOCK	YES	$5.00	$0.00	$5.00
Total										$3,888.63

Summary

- The Multi-Channel Tuner has been in successful operation since 2023 and provided the required radio circuit performance.
- Only 100 watts transmit power was necessary to achieve good performance.
- The unit was returned for lightning damage repairs which were completed on 24 SEP 2026 with the addition of the tuner HV relay disconnect and upgraded RCS-8V relays.
- While the MFJ components used are no longer available, there are replacement components that would allow a similar system at 150 watts to be built.

Questions?

Lewis Thompson – W5IFQ

Registered Professional Engineer

Personal Cell – 512-587-9944
Home E-mail: w5ifq@att.net